Reset Early, Reset Often, Eliminate Models

How aggressively resetting connection tableaux can saturate the SAT solver and find an unsatisfiability proof faster.

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Knights and Knaves



A traveller meets a husband and wife. He knows that either both of them are *knights*, who always tell the truth, or *knaves*, who always lie.

The husband makes a statement, and the traveller is given three clues.

1. If the husband's statement is false, then his wife is a knight.
2. If the wife is a knight, then the husband's statement is true.
3. If the husband is a knight, then his wife is a knight

Smullyan (1978): *What Is the Name of This Book?: The Riddle of Dracula and Other Logical Puzzles*. Englewood Cliffs, NJ: Prentice-Hall

PUZ020-1.p

$$a(\eta) \qquad \neg b(\eta)$$

$$\neg a(A) \vee \neg c\big(\phi(A)\big) \vee b(A)$$

$$c\big(\phi(\eta)\big) \vee b(\gamma) \qquad \neg b(\gamma) \vee c\big(\phi(\eta)\big)$$

$a(A)$ A is a person.

$b(A)$ A is a knight.

$c(B)$ B is a true statement.

$d(A)$ A is a knave.

$e(A, B)$ A says statement B.

$\phi(A)$ statement ϕ by A.

η the husband.

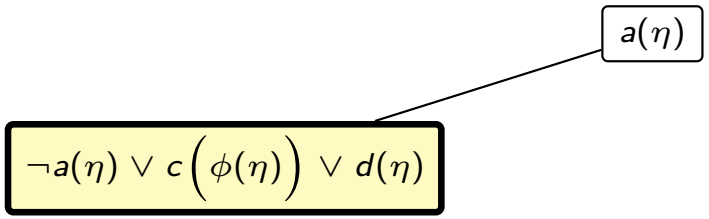
γ the wife.

Smullyan (1978): *What Is the Name of This Book?: The Riddle of Dracula and Other Logical Puzzles*. Englewood Cliffs, NJ: Prentice-Hall
Sutcliffe (2017): *The TPTP Problem Library and Associated Infrastructure: From CNF to TH0, TPTP v6.4.0*. *Journal of Automated Reasoning* 59(4)

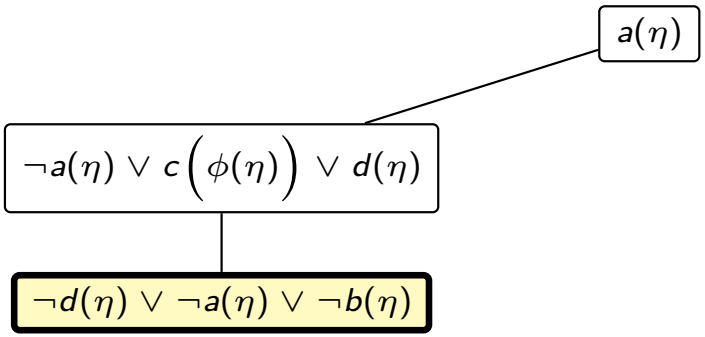
leanTAP

$a(\eta)$

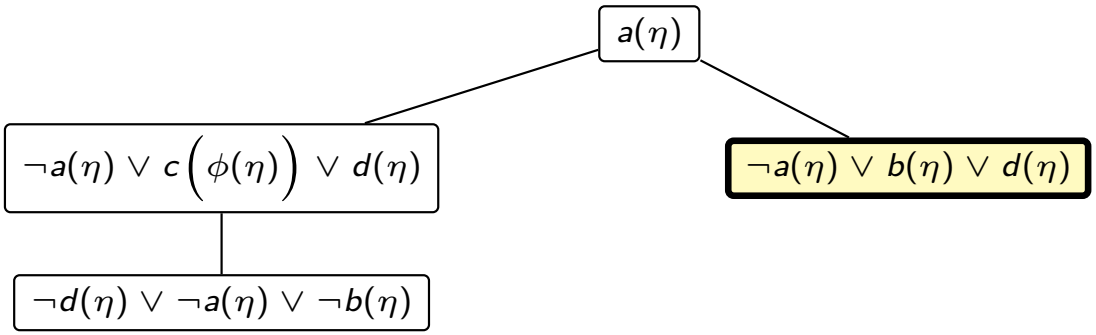
leanTAP



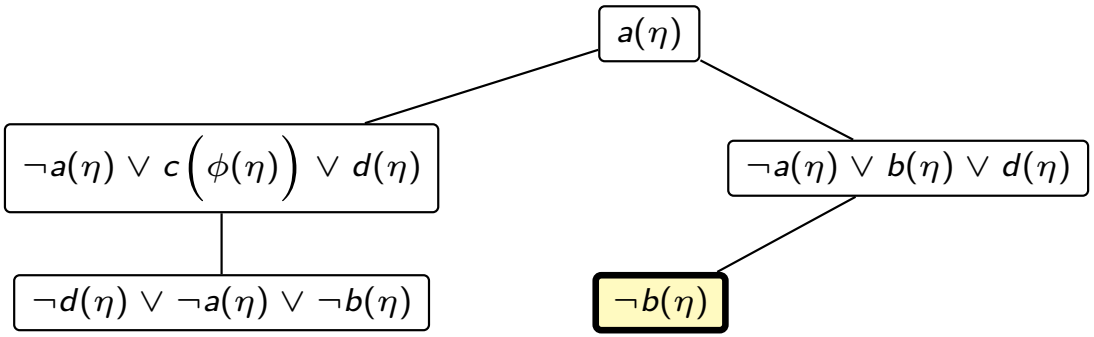
leanTAP


$a(\eta)$
$\neg a(\eta) \vee c\left(\phi(\eta)\right) \vee d(\eta)$
$\neg d(\eta) \vee \neg a(\eta) \vee \neg b(\eta)$

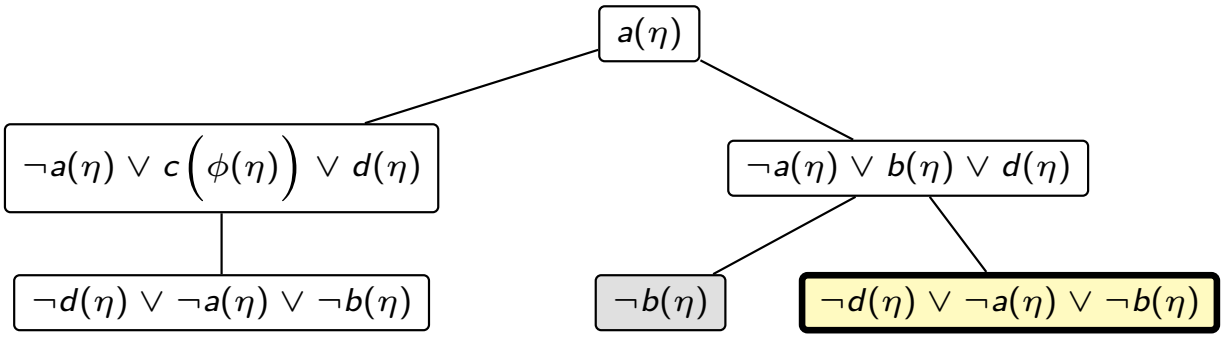
leanTAP


$a(\eta)$
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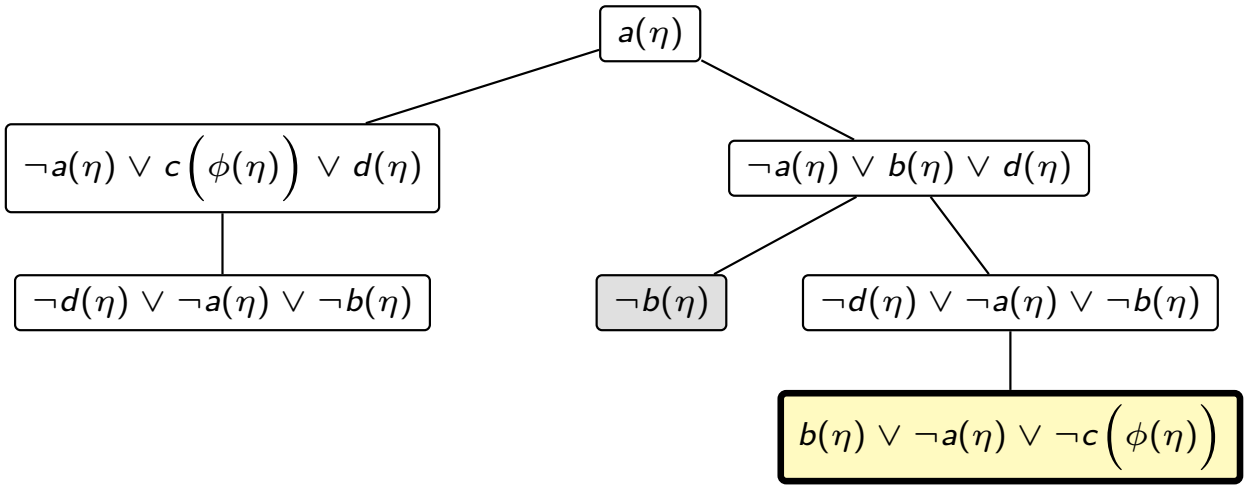
leanTAP


$a(\eta)$
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$\neg a(\eta) \vee b(\eta) \vee d(\eta)$
$\neg d(\eta) \vee \neg a(\eta) \vee \neg b(\eta)$
$\neg b(\eta)$

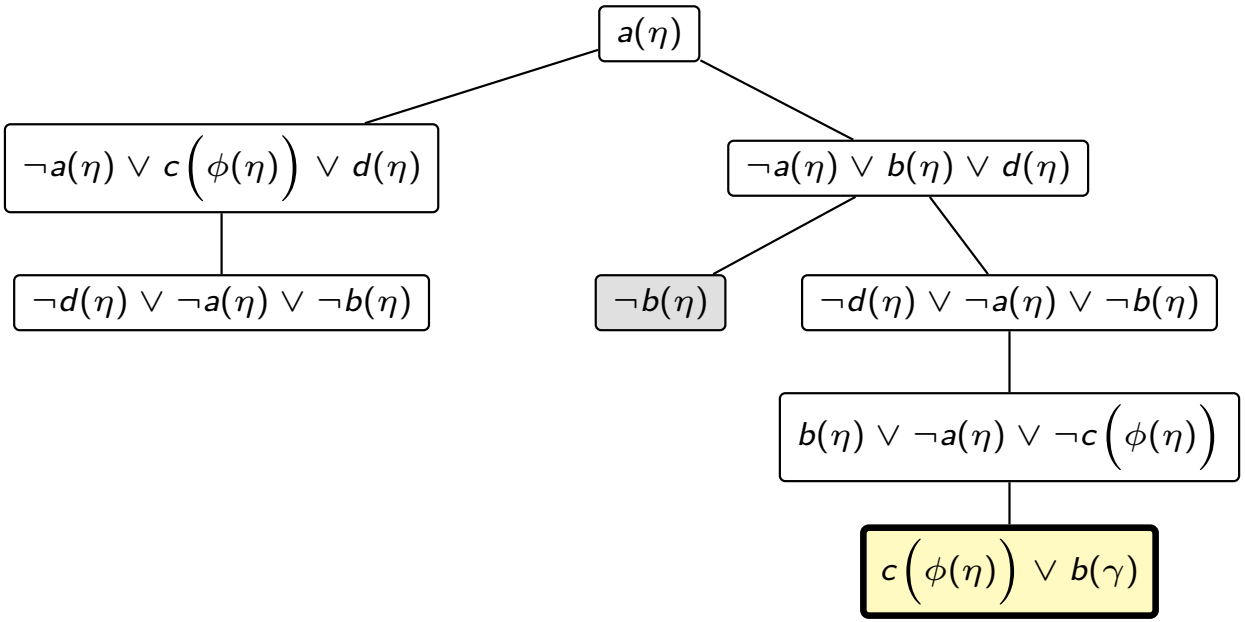
leanTAP


$a(\eta)$
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$\neg d(\eta) \vee \neg a(\eta) \vee \neg b(\eta)$
$\neg b(\eta)$
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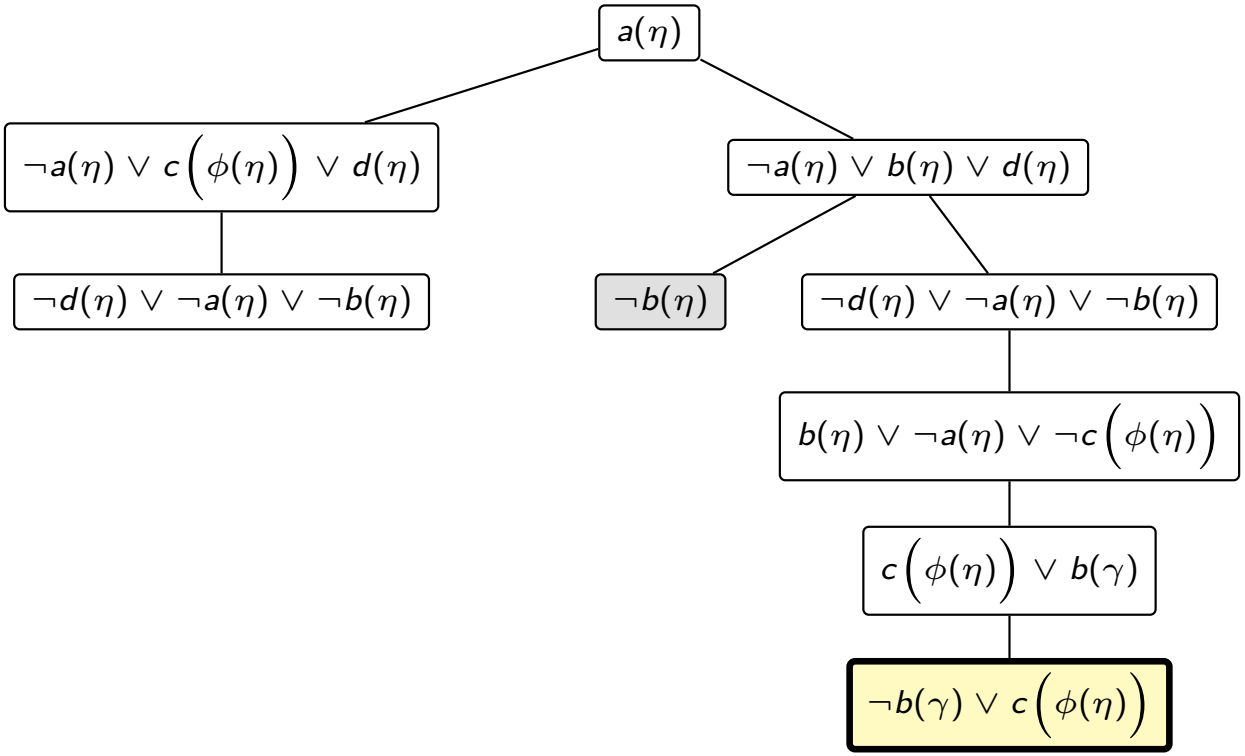
leanTAP


$a(\eta)$
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$\neg a(\eta) \vee b(\eta) \vee d(\eta)$
$\neg d(\eta) \vee \neg a(\eta) \vee \neg b(\eta)$
$\neg b(\eta)$
$\neg d(\eta) \vee \neg a(\eta) \vee \neg b(\eta)$
$b(\eta) \vee \neg a(\eta) \vee \neg c\left(\phi(\eta)\right)$

leanTAP


$a(\eta)$
$\neg a(\eta) \vee c\left(\phi(\eta)\right) \vee d(\eta)$
$\neg a(\eta) \vee b(\eta) \vee d(\eta)$
$\neg d(\eta) \vee \neg a(\eta) \vee \neg b(\eta)$
$\neg b(\eta)$
$\neg d(\eta) \vee \neg a(\eta) \vee \neg b(\eta)$
$b(\eta) \vee \neg a(\eta) \vee \neg c\left(\phi(\eta)\right)$
$c\left(\phi(\eta)\right) \vee b(\gamma)$

leanTAP



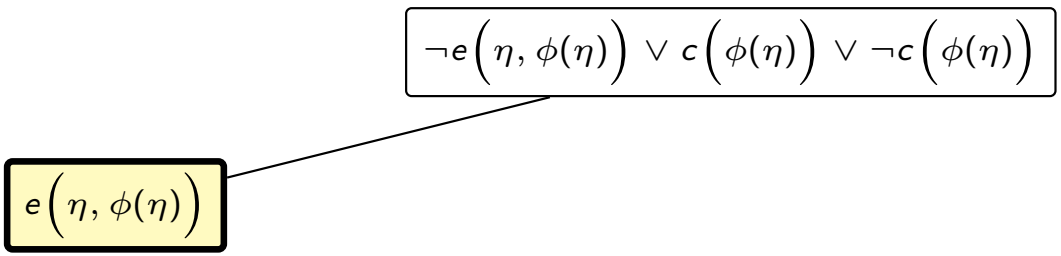
total steps: 2,563

leanCoP

$$\neg e\Big(\eta, \phi(\eta)\Big) \vee c\Big(\phi(\eta)\Big) \vee \neg c\Big(\phi(\eta)\Big)$$

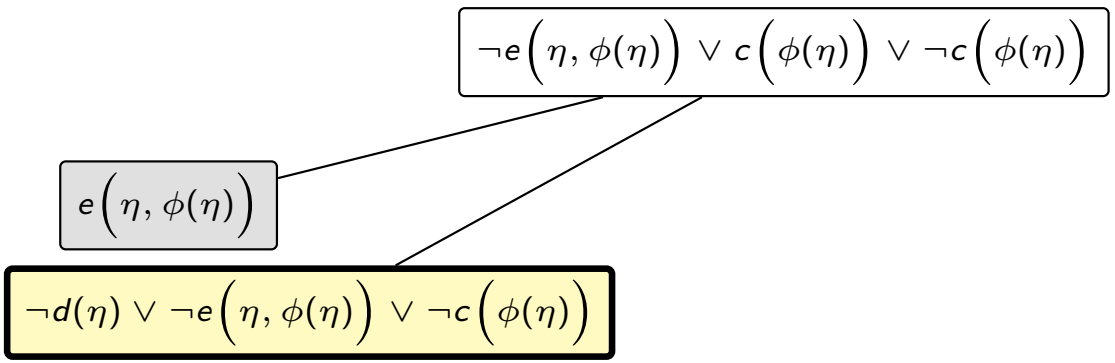

Otten and Bibel (2003): *leanCoP: Lean Connection-Based Theorem Proving*. *Journal of Symbolic Computation* 36(1–2)
Otten (2010): *Restricting Backtracking in Connection Calculi*. *AI Communications* 23(2–3)

leanCoP



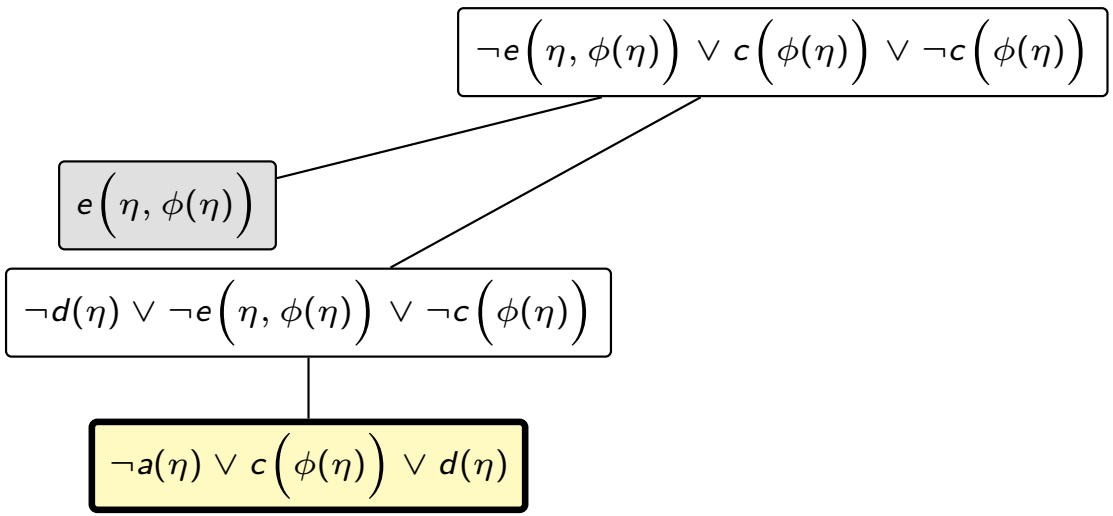
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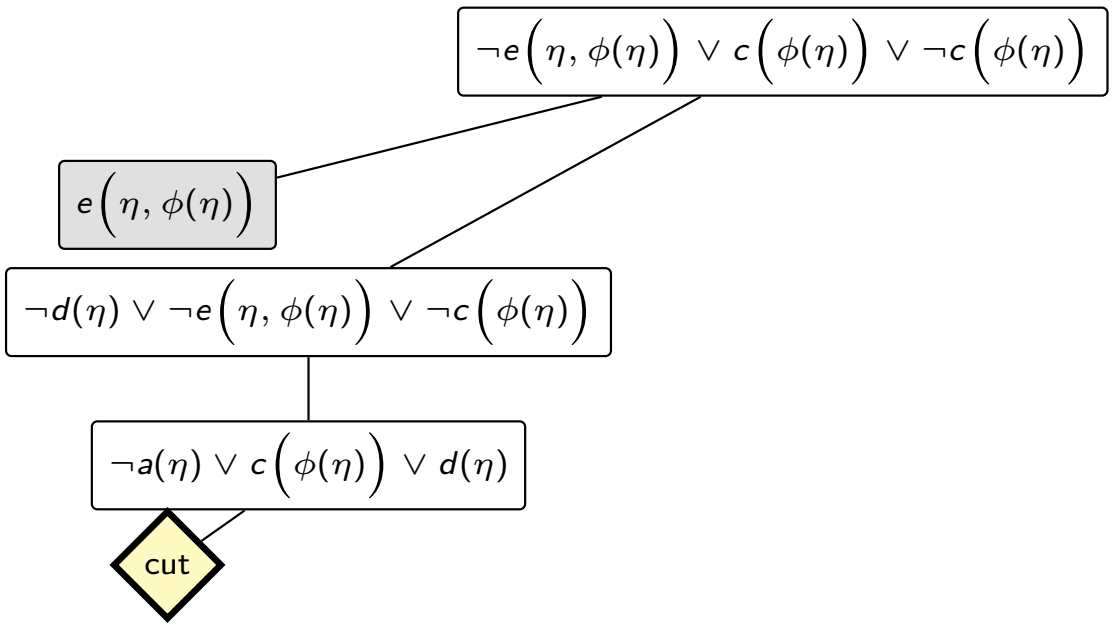
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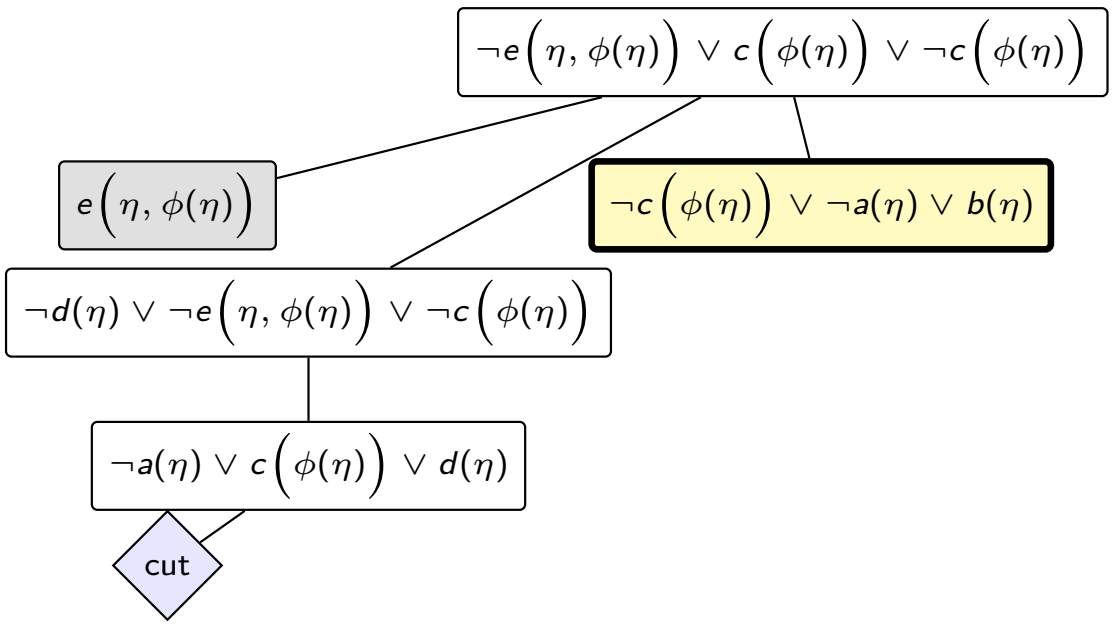
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leanCoP



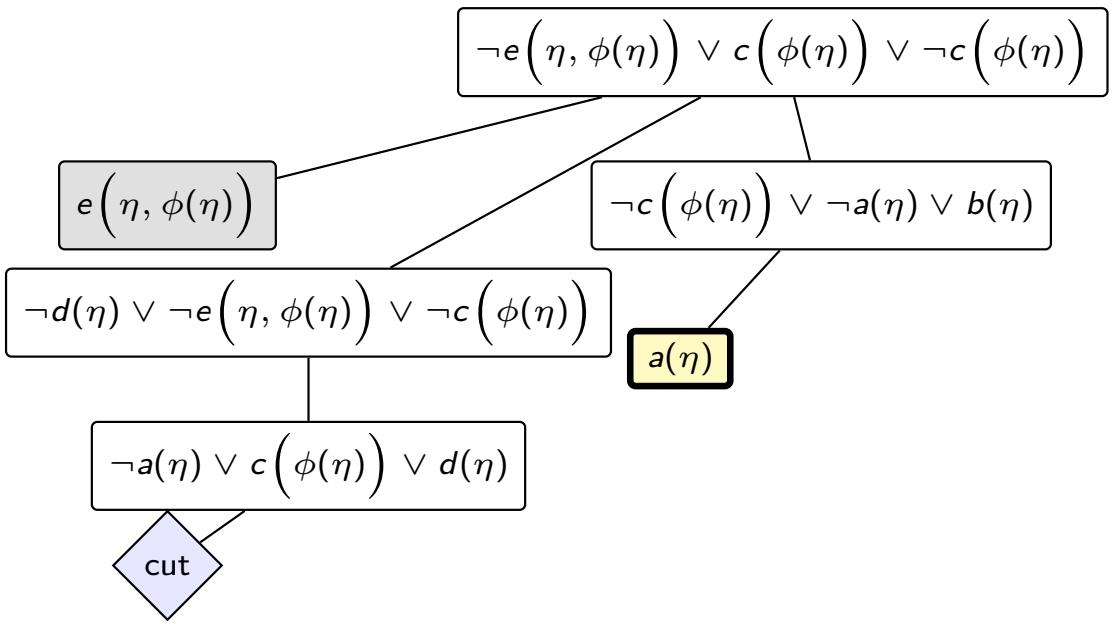
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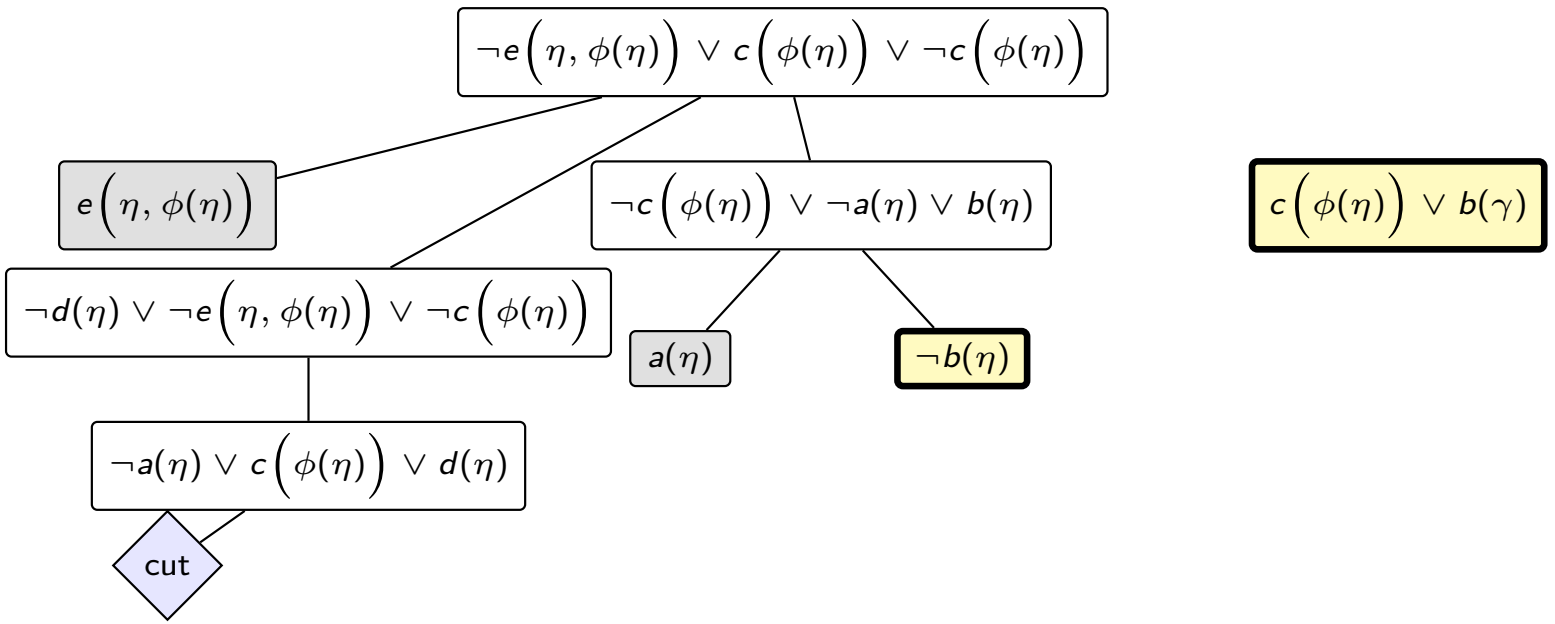
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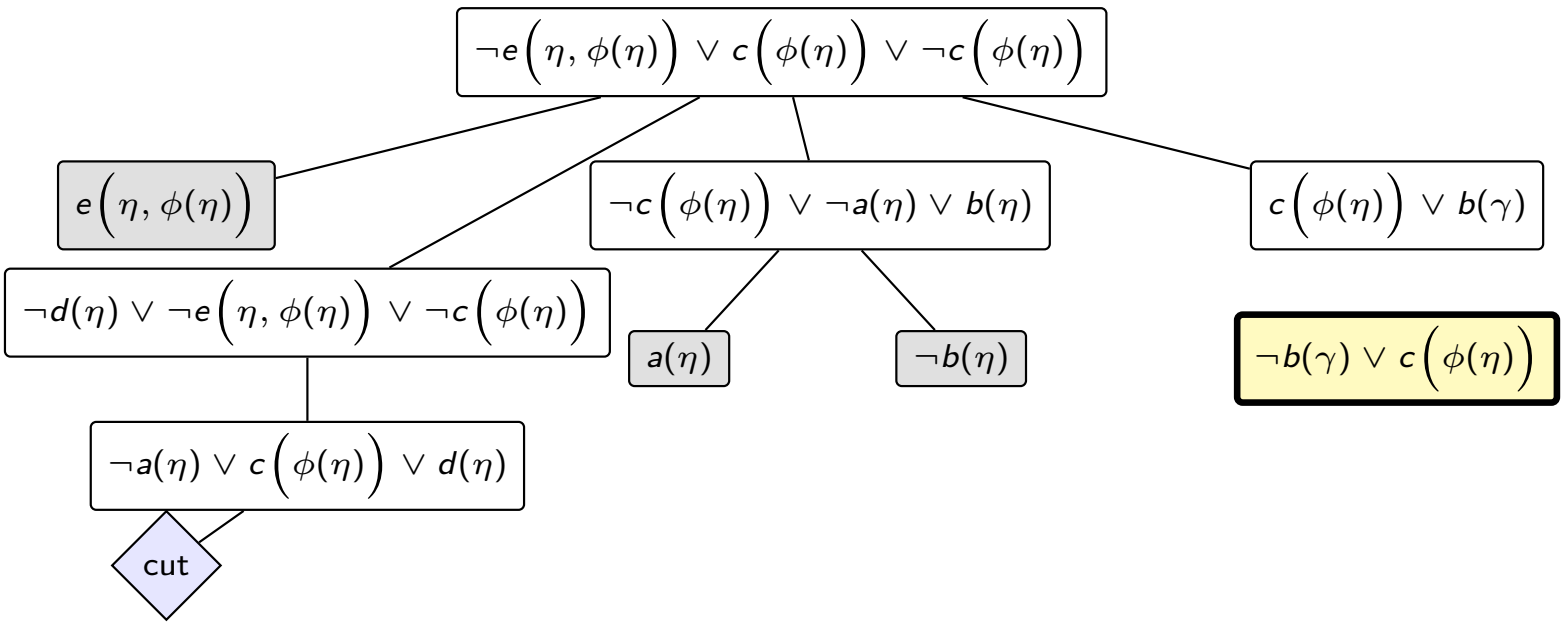
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leanCoP



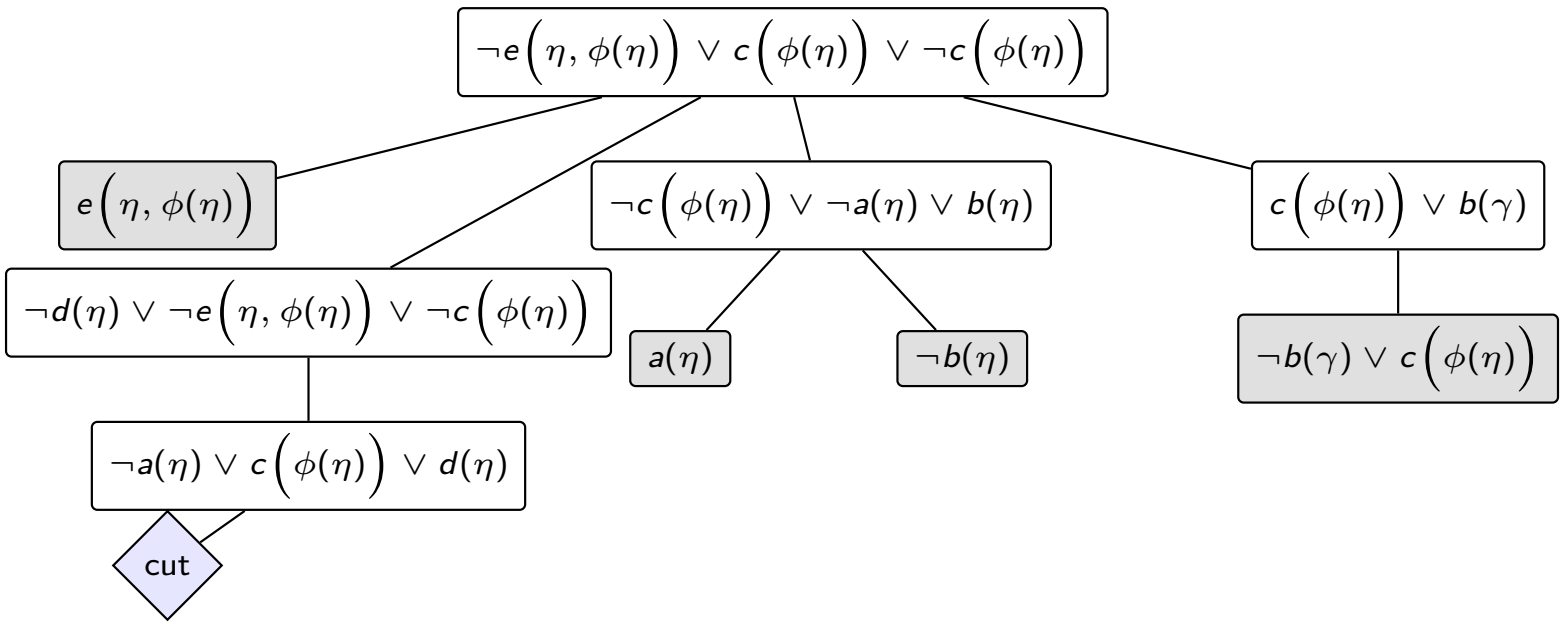
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leanCoP



total steps: 241

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SATCoP

$a(\eta)$

persistent SAT ground

$a(\eta)$

SATCoP

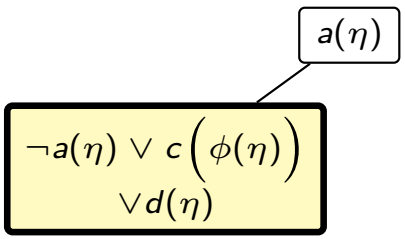
$a(\eta)$

persistent SAT ground

$a(\eta)$

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SATCoP

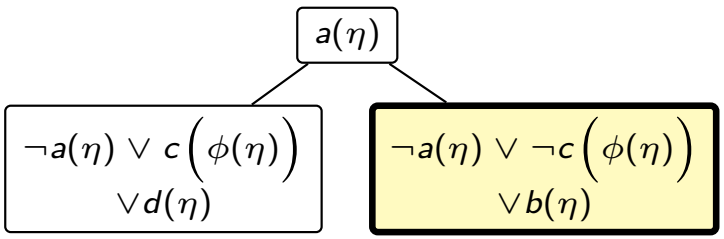


persistent SAT ground

$a(\eta)$

$\neg a(\eta) \vee c\Big(\phi(\eta)\Big) \vee d(\eta)$

SATCoP



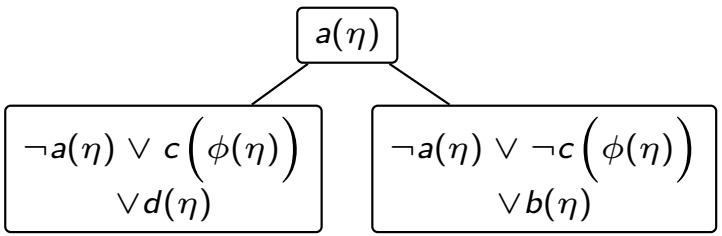
persistent SAT ground

$a(\eta)$

$\neg a(\eta) \vee c\left(\phi(\eta)\right) \vee d(\eta)$

$\neg a(\eta) \vee \neg c\left(\phi(\eta)\right) \vee b(\eta)$

SATCoP



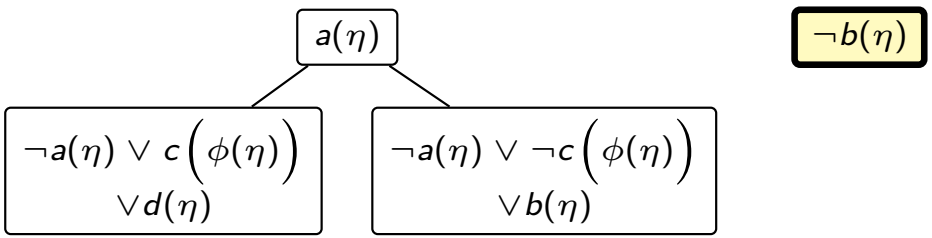
persistent SAT ground

$a(\eta)$

$\neg a(\eta) \vee c\Big(\phi(\eta)\Big) \vee d(\eta)$

$\neg a(\eta) \vee \neg c\Big(\phi(\eta)\Big) \vee b(\eta)$

SATCoP



$\neg b(\eta)$

persistent SAT ground

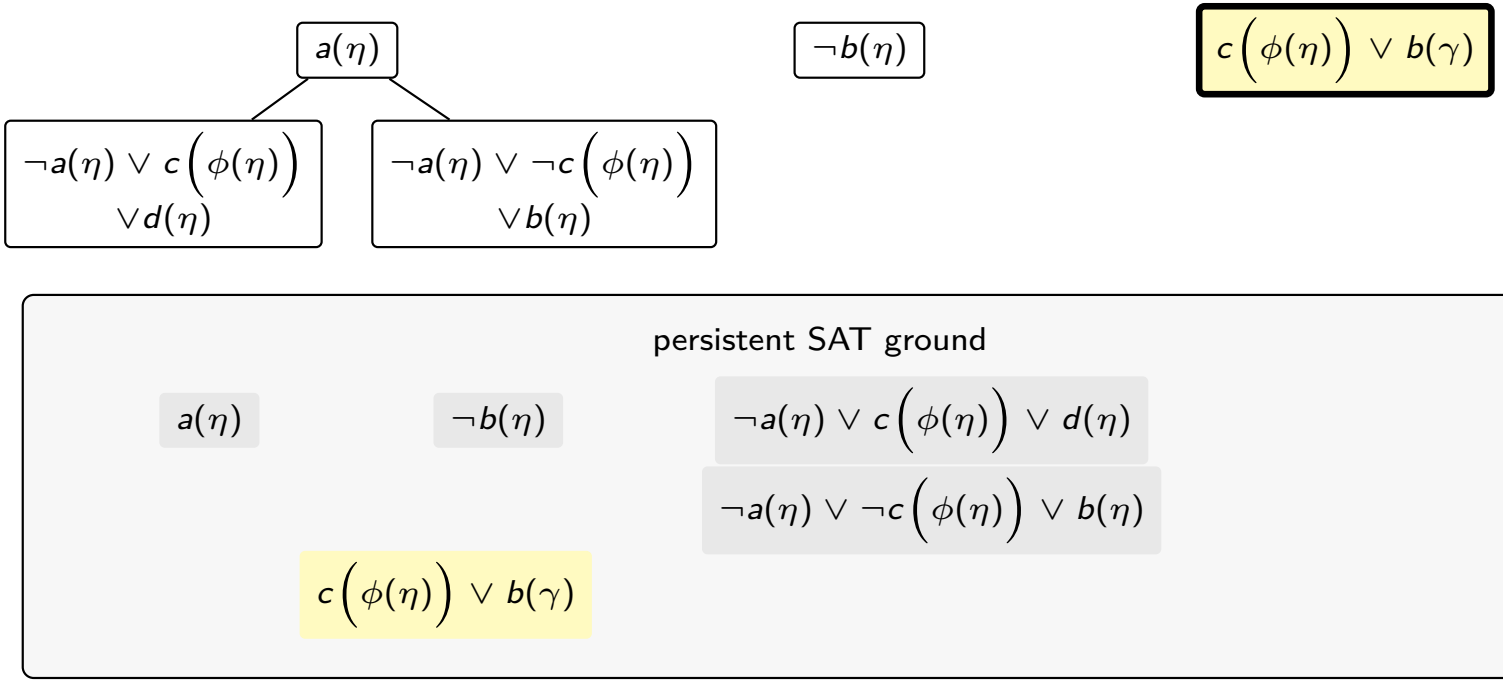
$a(\eta)$

$\neg b(\eta)$

$\neg a(\eta) \vee c\left(\phi(\eta)\right) \vee d(\eta)$

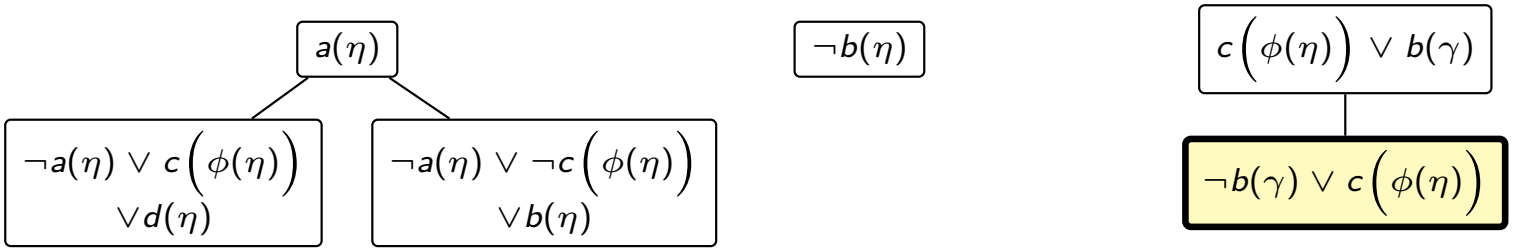
$\neg a(\eta) \vee \neg c\left(\phi(\eta)\right) \vee b(\eta)$

SATCoP


$a(\eta)$
$\neg b(\eta)$
$c\left(\phi(\eta)\right) \vee b(\gamma)$
$\neg a(\eta) \vee c\left(\phi(\eta)\right) \vee d(\eta)$
$\neg a(\eta) \vee \neg c\left(\phi(\eta)\right) \vee b(\eta)$
persistent SAT ground
$a(\eta)$
$\neg b(\eta)$
$\neg a(\eta) \vee c\left(\phi(\eta)\right) \vee d(\eta)$
$\neg a(\eta) \vee \neg c\left(\phi(\eta)\right) \vee b(\eta)$
$c\left(\phi(\eta)\right) \vee b(\gamma)$

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SATCoP



persistent SAT ground

$a(\eta)$

$\neg b(\eta)$

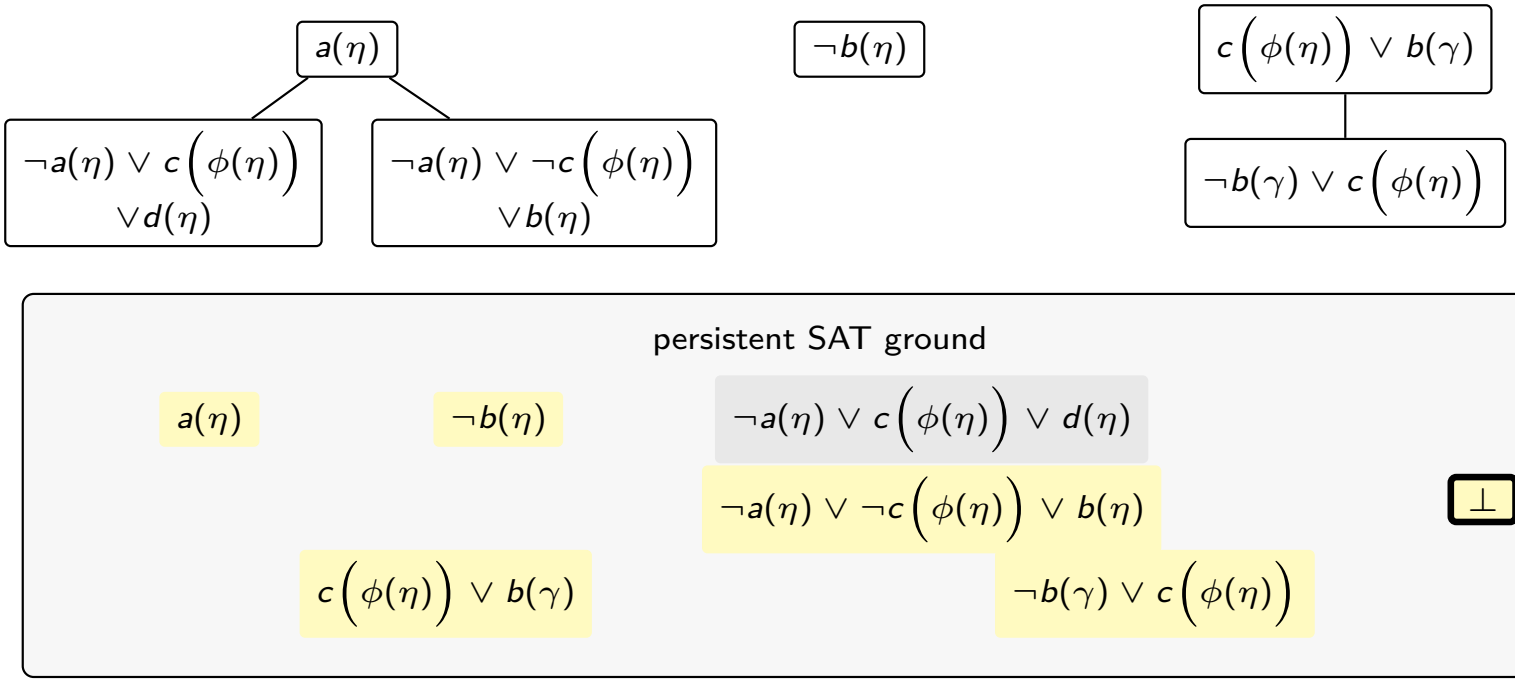
$\neg a(\eta) \vee c\left(\phi(\eta)\right) \vee d(\eta)$

$\neg a(\eta) \vee \neg c\left(\phi(\eta)\right) \vee b(\eta)$

$c\left(\phi(\eta)\right) \vee b(\gamma)$

$\neg b(\gamma) \vee c\left(\phi(\eta)\right)$

SATCoP



total steps: 93

- **SATCoP$_0$** uses the SAT solver to check whether the ground clauses are inconsistent.
- **SATCoP** also looks at the current SAT model.
- When choosing the next connection, it tries a goal that is true in that model. If the model gives no preference, it keeps the shuffled order.

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SATCoP

$$c\Big(\phi(\eta)\Big) \vee b(\gamma)$$

persistent SAT ground

$$c\Big(\phi(\eta)\Big) \vee b(\gamma)$$

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SATCoP

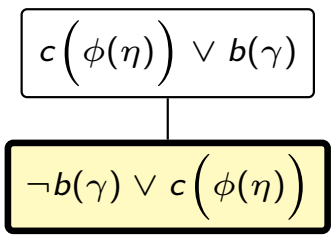
$$c\Big(\phi(\eta)\Big) \vee b(\gamma)$$

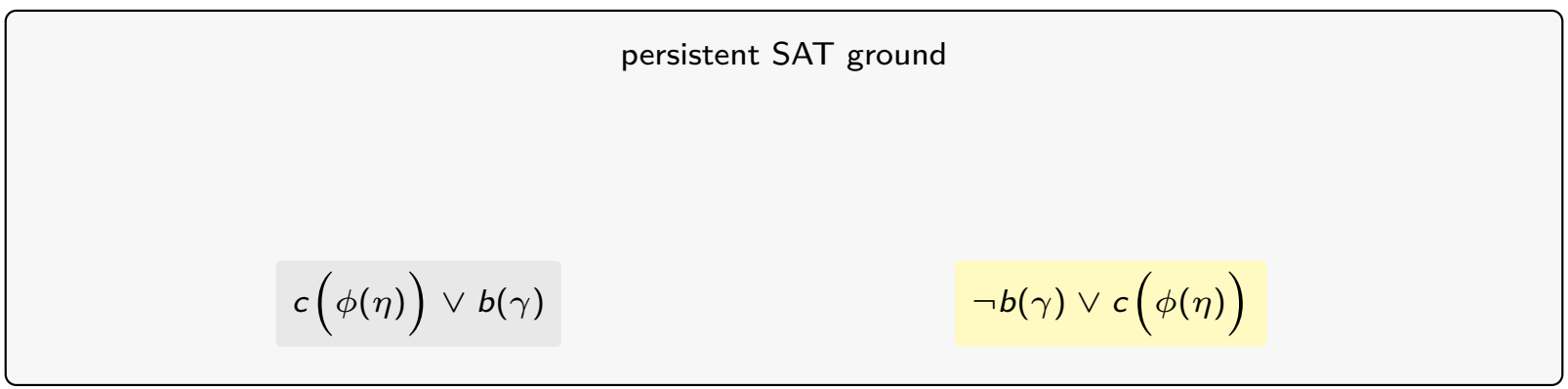
persistent SAT ground

$$c\Big(\phi(\eta)\Big) \vee b(\gamma)$$

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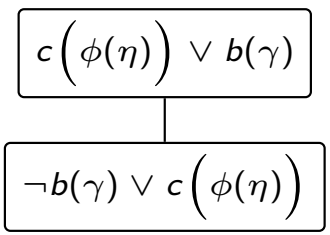
SATCoP

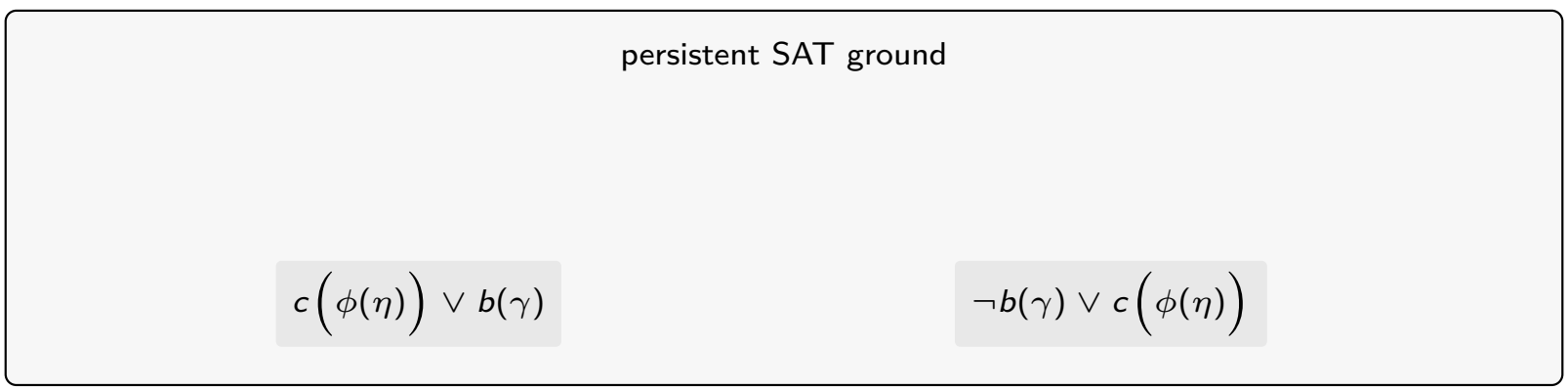




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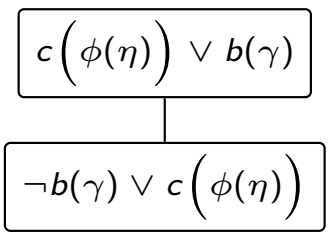
SATCoP





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SATCoP



$a(\eta)$

persistent SAT ground

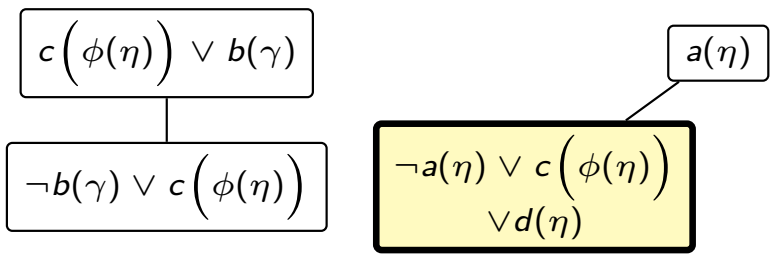
$a(\eta)$

$c\left(\phi(\eta)\right) \vee b(\gamma)$

$\neg b(\gamma) \vee c\left(\phi(\eta)\right)$

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SATCoP



persistent SAT ground

$a(\eta)$

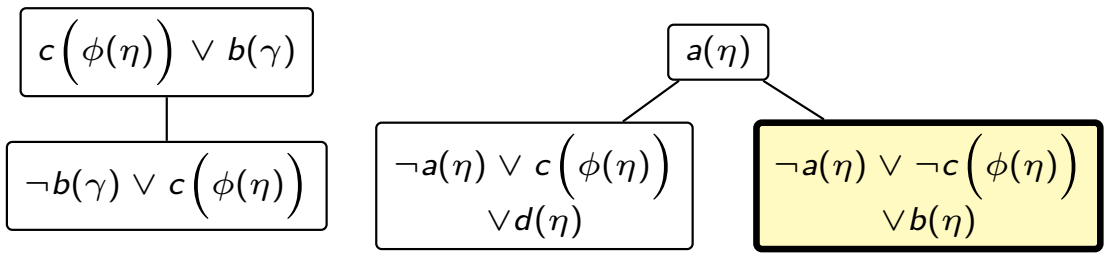
$\neg a(\eta) \vee c\left(\phi(\eta)\right) \vee d(\eta)$

$c\left(\phi(\eta)\right) \vee b(\gamma)$

$\neg b(\gamma) \vee c\left(\phi(\eta)\right)$

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SATCoP



persistent SAT ground

$a(\eta)$

$\neg a(\eta) \vee c\left(\phi(\eta)\right) \vee d(\eta)$

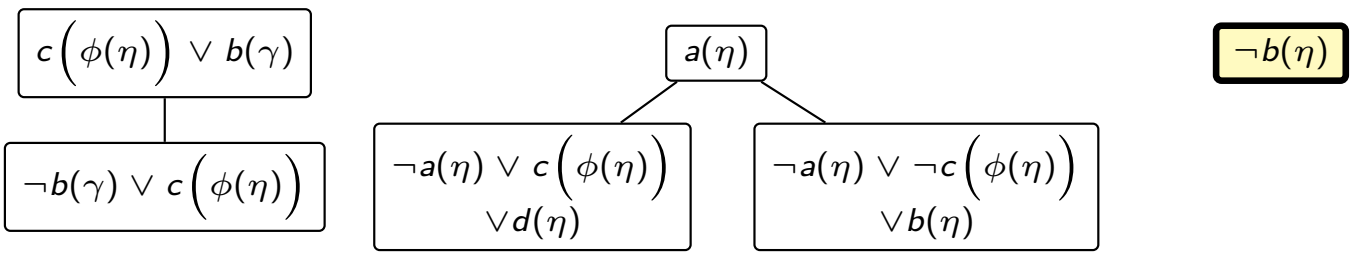
$\neg a(\eta) \vee \neg c\left(\phi(\eta)\right) \vee b(\eta)$

$c\left(\phi(\eta)\right) \vee b(\gamma)$

$\neg b(\gamma) \vee c\left(\phi(\eta)\right)$

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SATCoP



persistent SAT ground

$a(\eta)$

$\neg b(\eta)$

$\neg a(\eta) \vee c\left(\phi(\eta)\right) \vee d(\eta)$

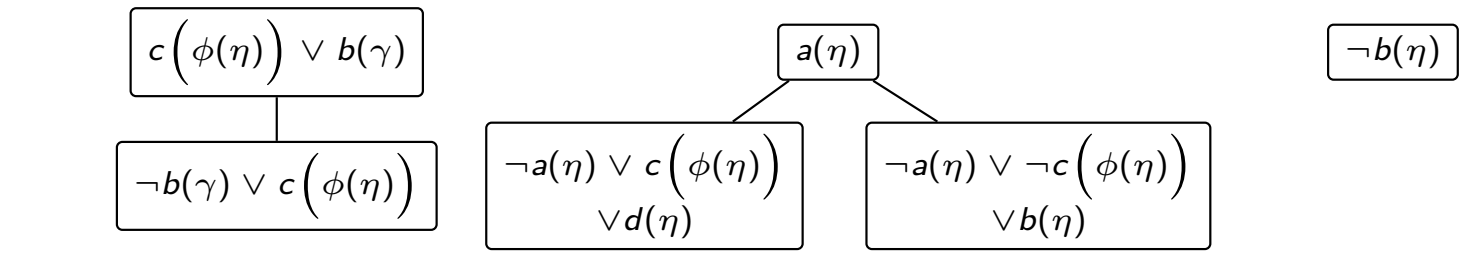
$\neg a(\eta) \vee \neg c\left(\phi(\eta)\right) \vee b(\eta)$

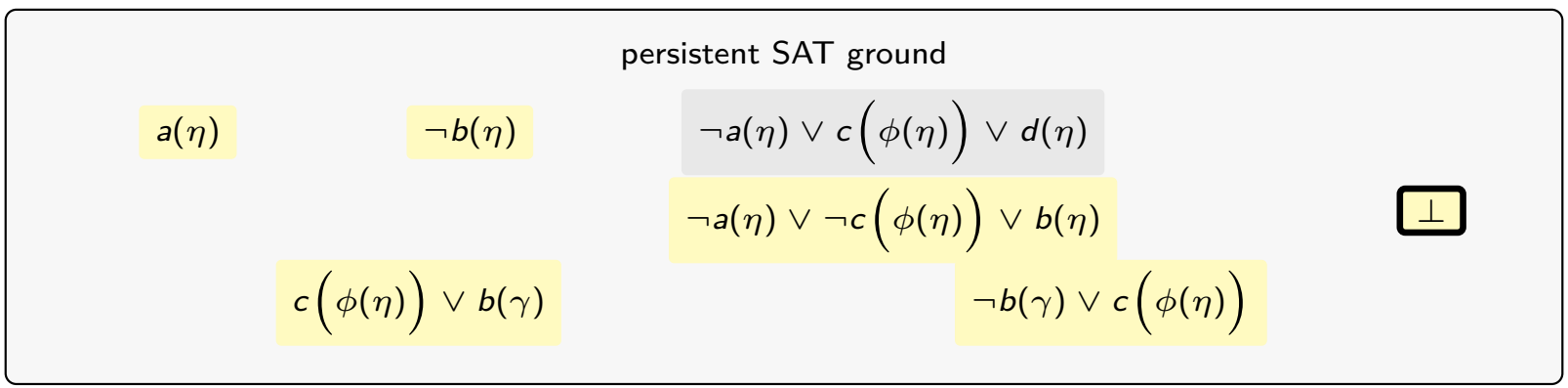
$c\left(\phi(\eta)\right) \vee b(\gamma)$

$\neg b(\gamma) \vee c\left(\phi(\eta)\right)$

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SATCoP





total steps: 75

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Connection tableau

Performs goal-directed search for relevant ground instances.

+

Persistent SAT solver

Retains those instances and detects propositional **UNSAT**.

A failed tableau can still contribute to the proof.

- Keep newly generated ground clauses.
- **Reset** the tableau instead of locally backtracking.
- Increase the depth only when an attempt produces nothing new.

SAT refutations remain **sound**; the reset search is **incomplete**.

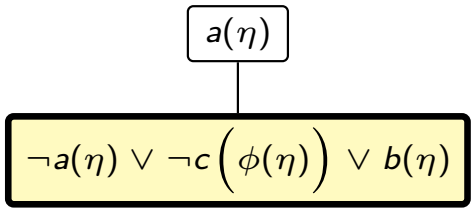
SATResetCoP

$a(\eta)$

persistent SAT ground

$a(\eta)$

SATResetCoP

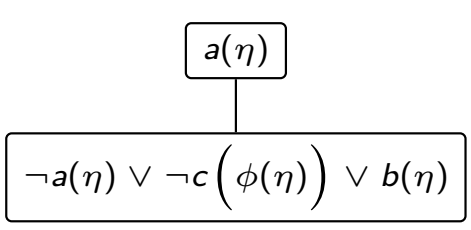


persistent SAT ground

$a(\eta)$

$\neg a(\eta) \vee \neg c\Big(\phi(\eta)\Big) \vee b(\eta)$

SATResetCoP



$$c\Big(\phi(\eta)\Big) \vee b(\gamma)$$

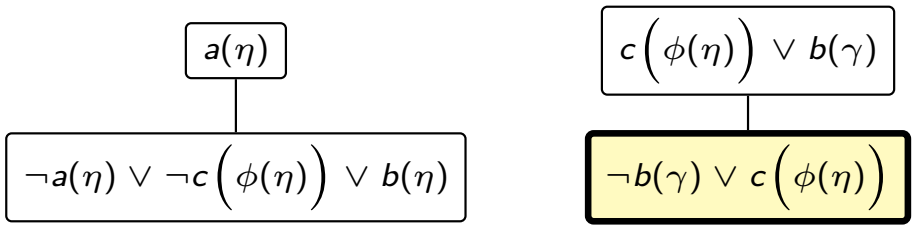
persistent SAT ground

$a(\eta)$

$$\neg a(\eta) \vee \neg c\Big(\phi(\eta)\Big) \vee b(\eta)$$

$$c\Big(\phi(\eta)\Big) \vee b(\gamma)$$

SATResetCoP



persistent SAT ground

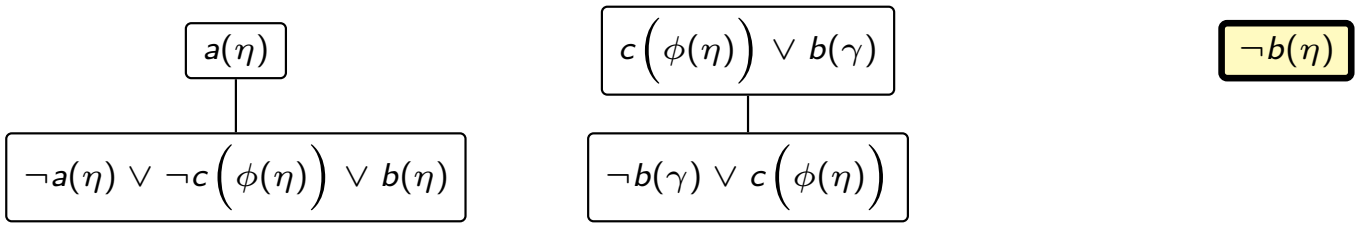
$a(\eta)$

$\neg a(\eta) \vee \neg c\left(\phi(\eta)\right) \vee b(\eta)$

$c\left(\phi(\eta)\right) \vee b(\gamma)$

$\neg b(\gamma) \vee c\left(\phi(\eta)\right)$

SATResetCoP



persistent SAT ground

$a(\eta)$

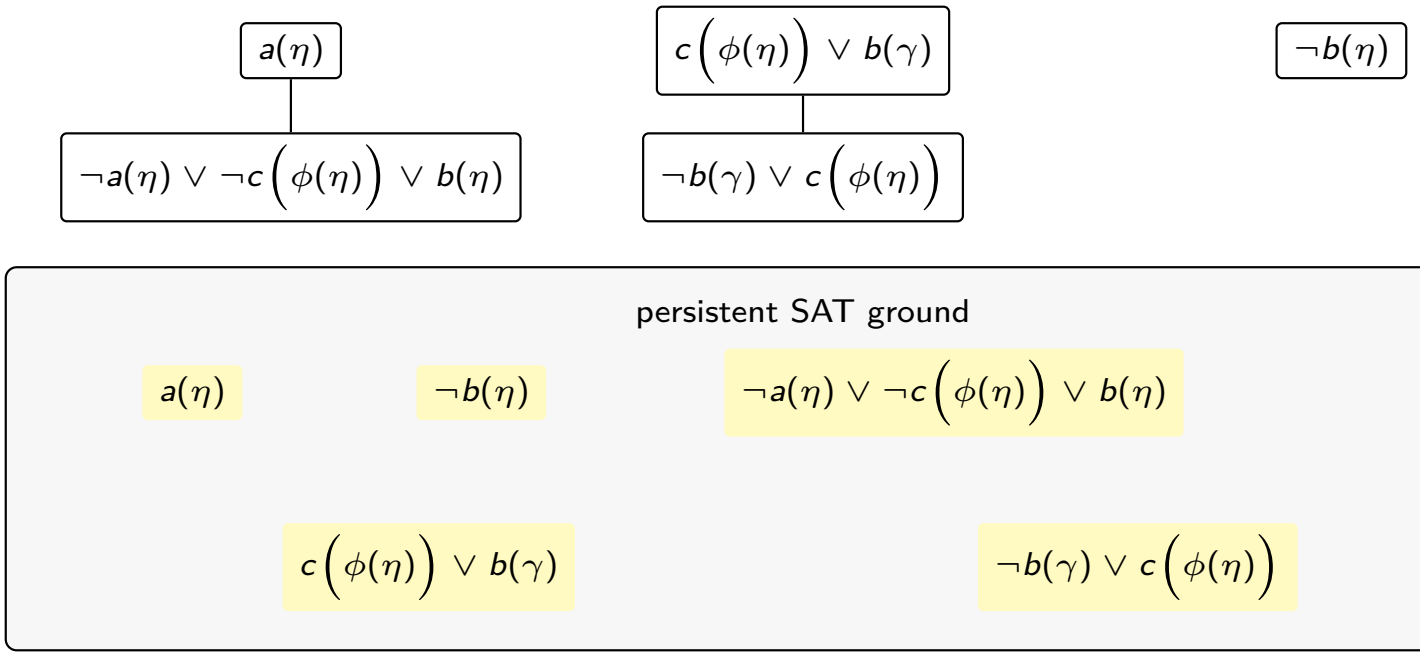
$\neg b(\eta)$

$\neg a(\eta) \vee \neg c\left(\phi(\eta)\right) \vee b(\eta)$

$c\left(\phi(\eta)\right) \vee b(\gamma)$

$\neg b(\gamma) \vee c\left(\phi(\eta)\right)$

SATResetCoP



total steps: 10

SYN081-1.p

$$\neg p(X) \vee \neg p(\psi(X))$$
$$p(\psi(X)) \vee p(X)$$
$$\neg p(X) \vee p(\psi(X))$$

$p(X)$ an arbitrary unary atom.

$\psi(X)$ an arbitrary unary function.

X a variable, universally quantified in every clause.

κ the fresh constant used to ground clauses for the SAT shadow.

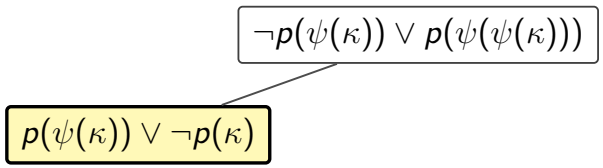
SATCoP

$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$

persistent SAT ground

$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$

SATCoP

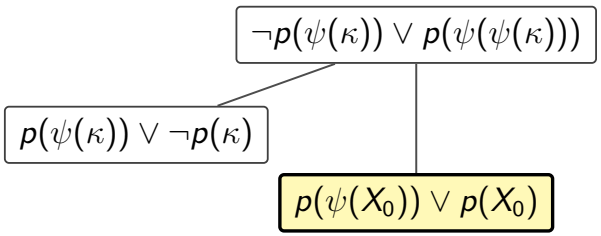


persistent SAT ground

$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$

$p(\psi(\kappa)) \vee \neg p(\kappa)$

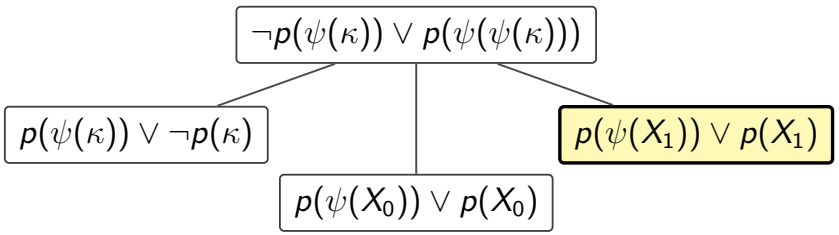
SATCoP



persistent SAT ground

$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$ $p(\psi(\kappa)) \vee \neg p(\kappa)$ $p(\psi(\kappa)) \vee p(\kappa)$

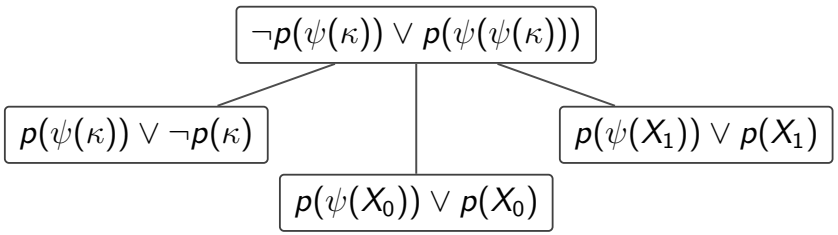
SATCoP



persistent SAT ground

$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$ $p(\psi(\kappa)) \vee \neg p(\kappa)$ $p(\psi(\kappa)) \vee p(\kappa)$

SATCoP

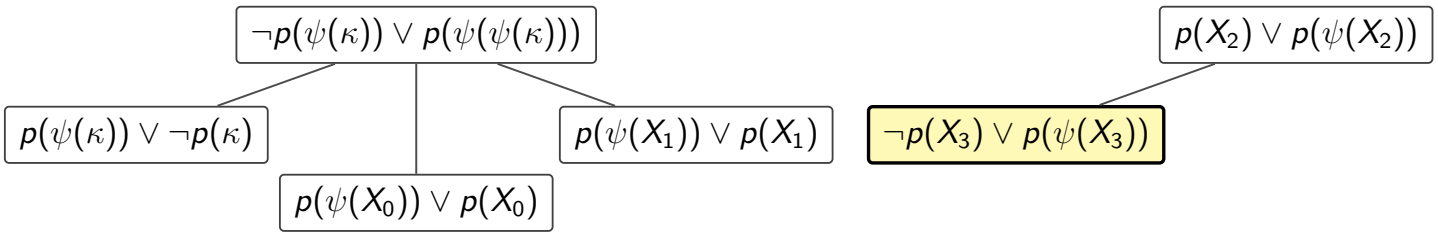


$p(X_2) \vee p(\psi(X_2))$

persistent SAT ground

$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$ $p(\psi(\kappa)) \vee \neg p(\kappa)$ $p(\psi(\kappa)) \vee p(\kappa)$

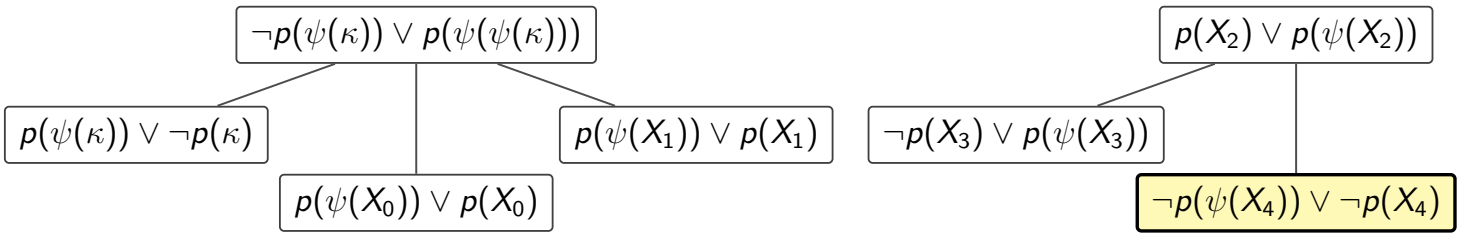
SATCoP



persistent SAT ground

$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$ $p(\psi(\kappa)) \vee \neg p(\kappa)$ $p(\psi(\kappa)) \vee p(\kappa)$

SATCoP



persistent SAT ground

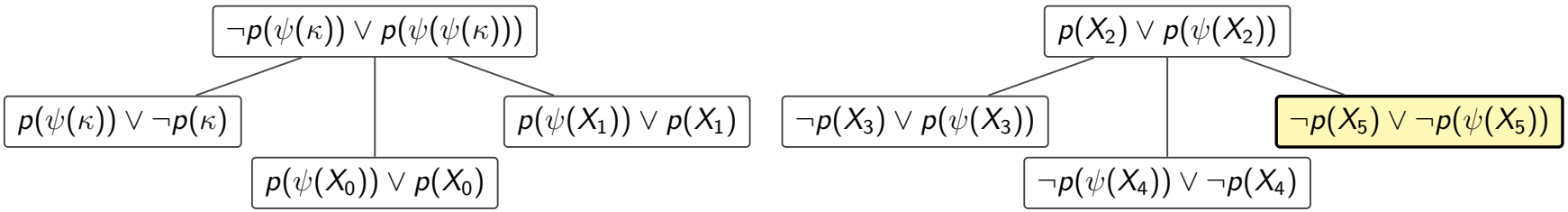
$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$

$p(\psi(\kappa)) \vee \neg p(\kappa)$

$p(\psi(\kappa)) \vee p(\kappa)$

$\neg p(\psi(\kappa)) \vee \neg p(\kappa)$

SATCoP



persistent SAT ground

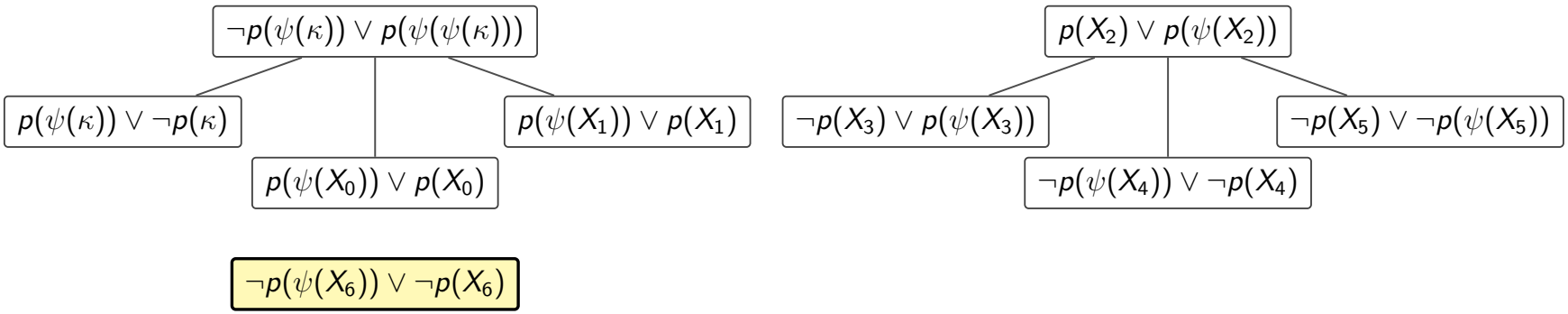
$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$

$p(\psi(\kappa)) \vee \neg p(\kappa)$

$p(\psi(\kappa)) \vee p(\kappa)$

$\neg p(\psi(\kappa)) \vee \neg p(\kappa)$

SATCoP



persistent SAT ground

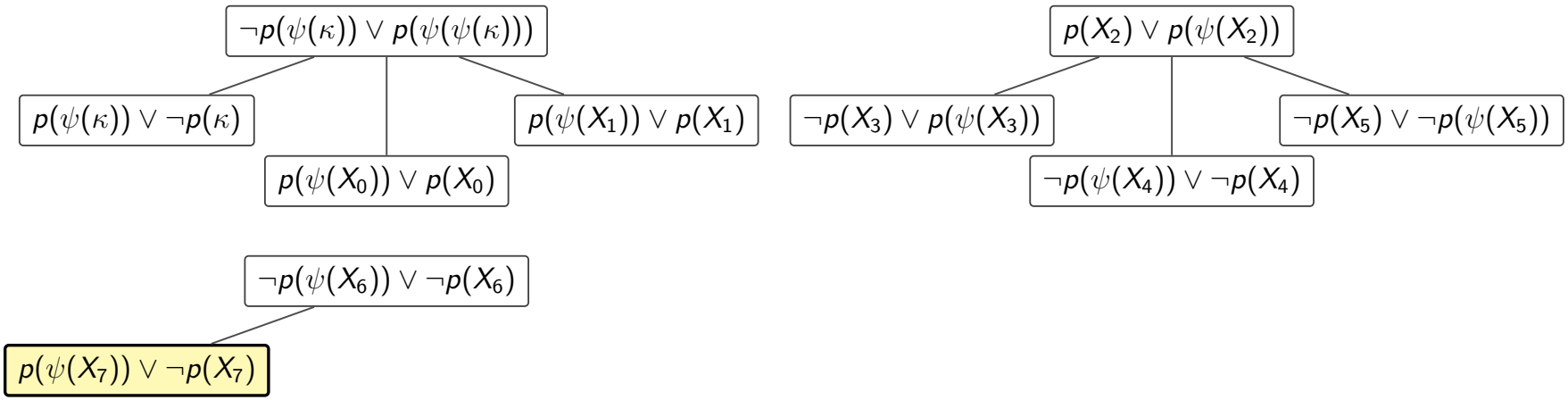
$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$

$p(\psi(\kappa)) \vee \neg p(\kappa)$

$p(\psi(\kappa)) \vee p(\kappa)$

$\neg p(\psi(\kappa)) \vee \neg p(\kappa)$

SATCoP

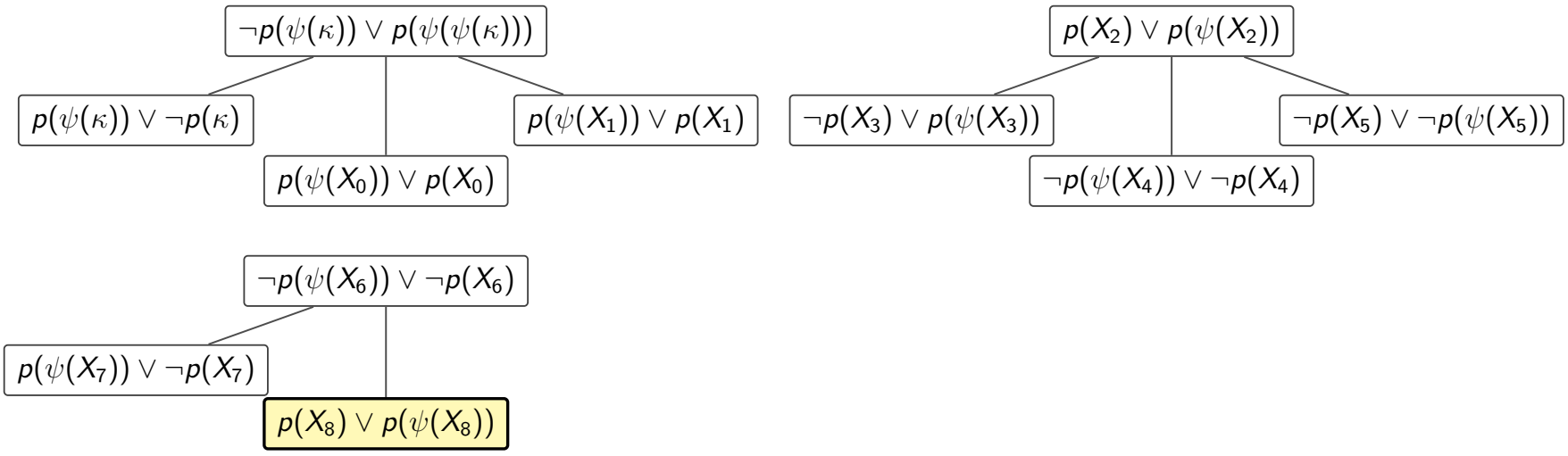


persistent SAT ground

$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$ $p(\psi(\kappa)) \vee \neg p(\kappa)$ $p(\psi(\kappa)) \vee p(\kappa)$

$\neg p(\psi(\kappa)) \vee \neg p(\kappa)$

SATCoP

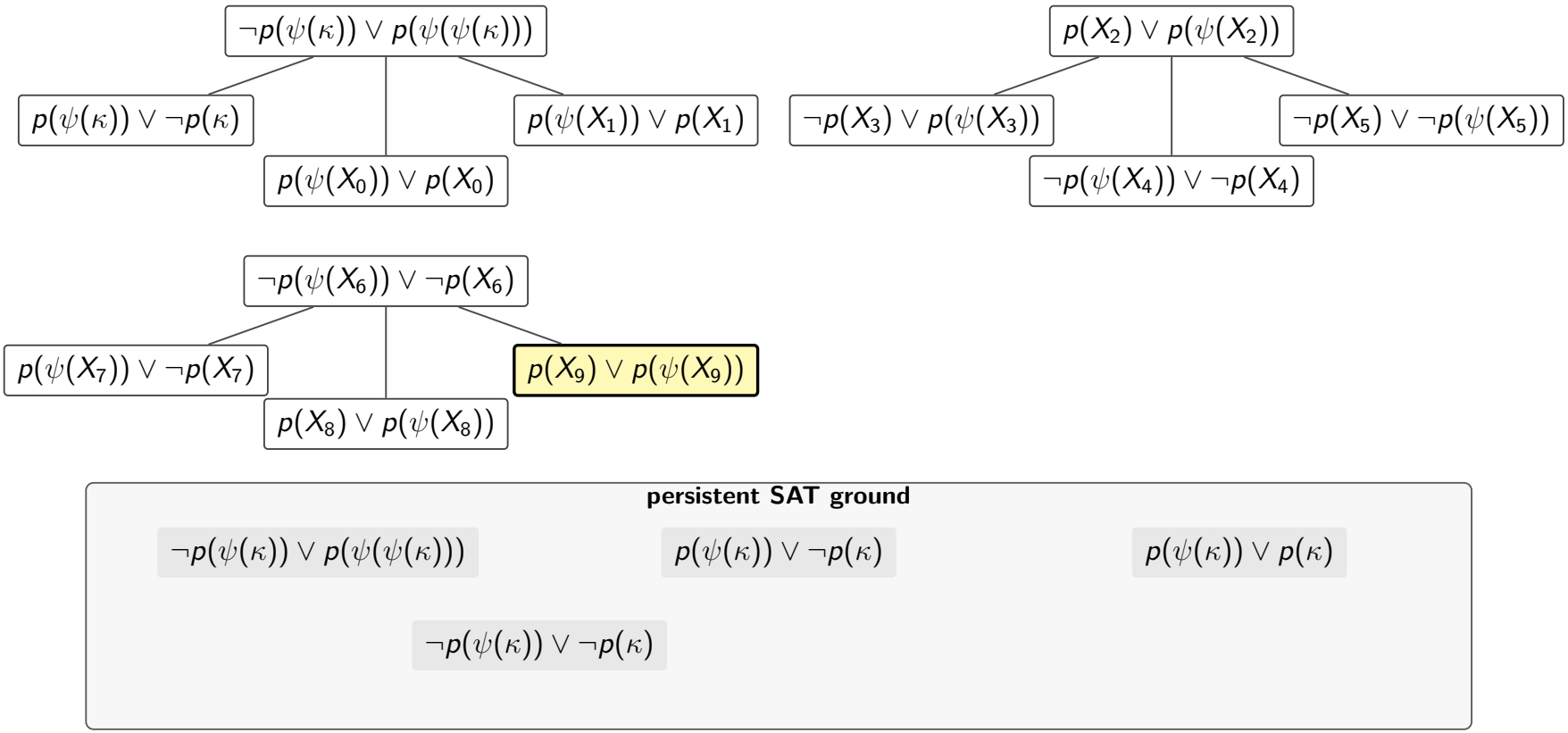


persistent SAT ground

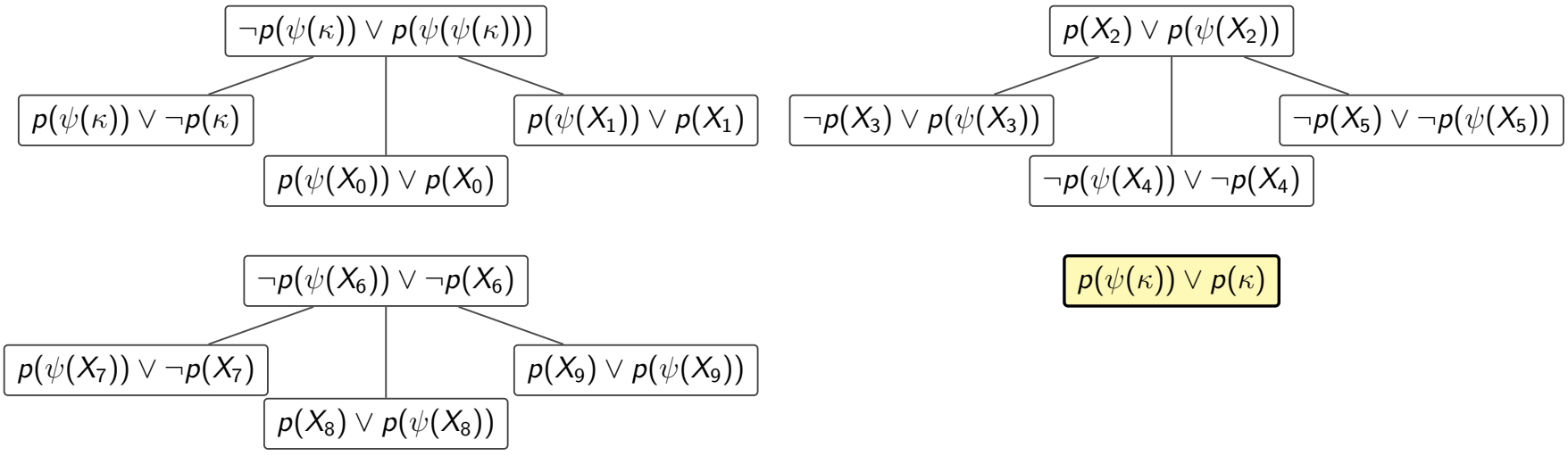
$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$ $p(\psi(\kappa)) \vee \neg p(\kappa)$ $p(\psi(\kappa)) \vee p(\kappa)$

$\neg p(\psi(\kappa)) \vee \neg p(\kappa)$

SATCoP


$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$
$p(\psi(\kappa)) \vee \neg p(\kappa)$
$p(\psi(X_1)) \vee p(X_1)$
$p(\psi(X_0)) \vee p(X_0)$
$p(X_2) \vee p(\psi(X_2))$
$\neg p(X_3) \vee p(\psi(X_3))$
$\neg p(X_5) \vee \neg p(\psi(X_5))$
$\neg p(\psi(X_4)) \vee \neg p(X_4)$
$\neg p(\psi(X_6)) \vee \neg p(X_6)$
$p(\psi(X_7)) \vee \neg p(X_7)$
$p(X_9) \vee p(\psi(X_9))$
$p(X_8) \vee p(\psi(X_8))$
persistent SAT ground
$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$
$p(\psi(\kappa)) \vee \neg p(\kappa)$
$p(\psi(\kappa)) \vee p(\kappa)$
$\neg p(\psi(\kappa)) \vee \neg p(\kappa)$

SATCoP

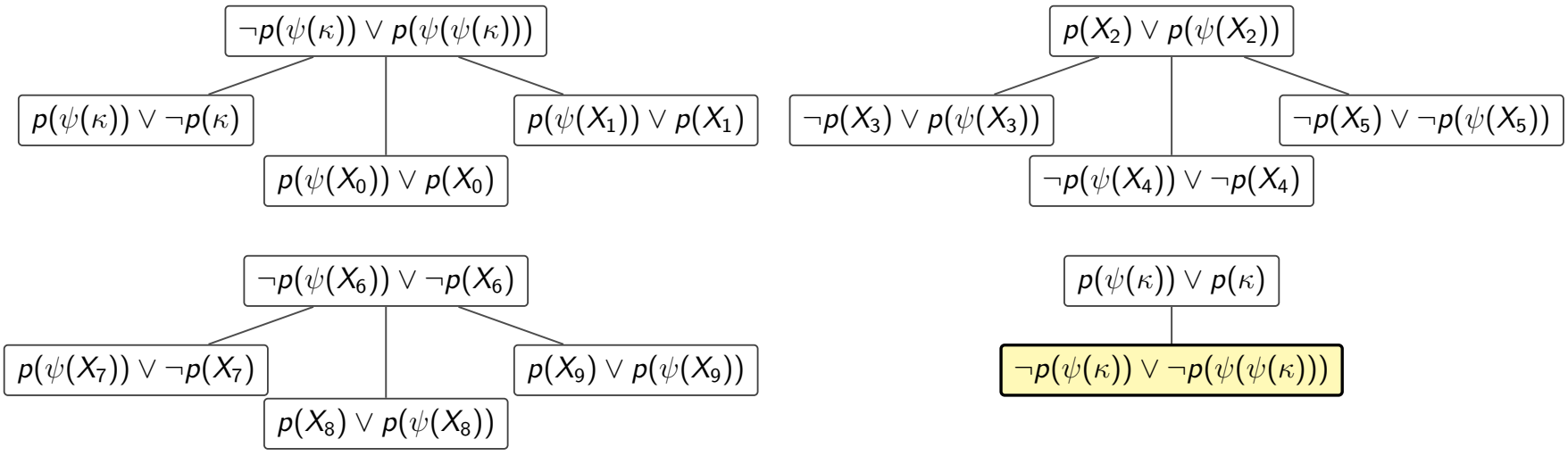


persistent SAT ground

$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$ $p(\psi(\kappa)) \vee \neg p(\kappa)$ $p(\psi(\kappa)) \vee p(\kappa)$

$\neg p(\psi(\kappa)) \vee \neg p(\kappa)$

SATCoP

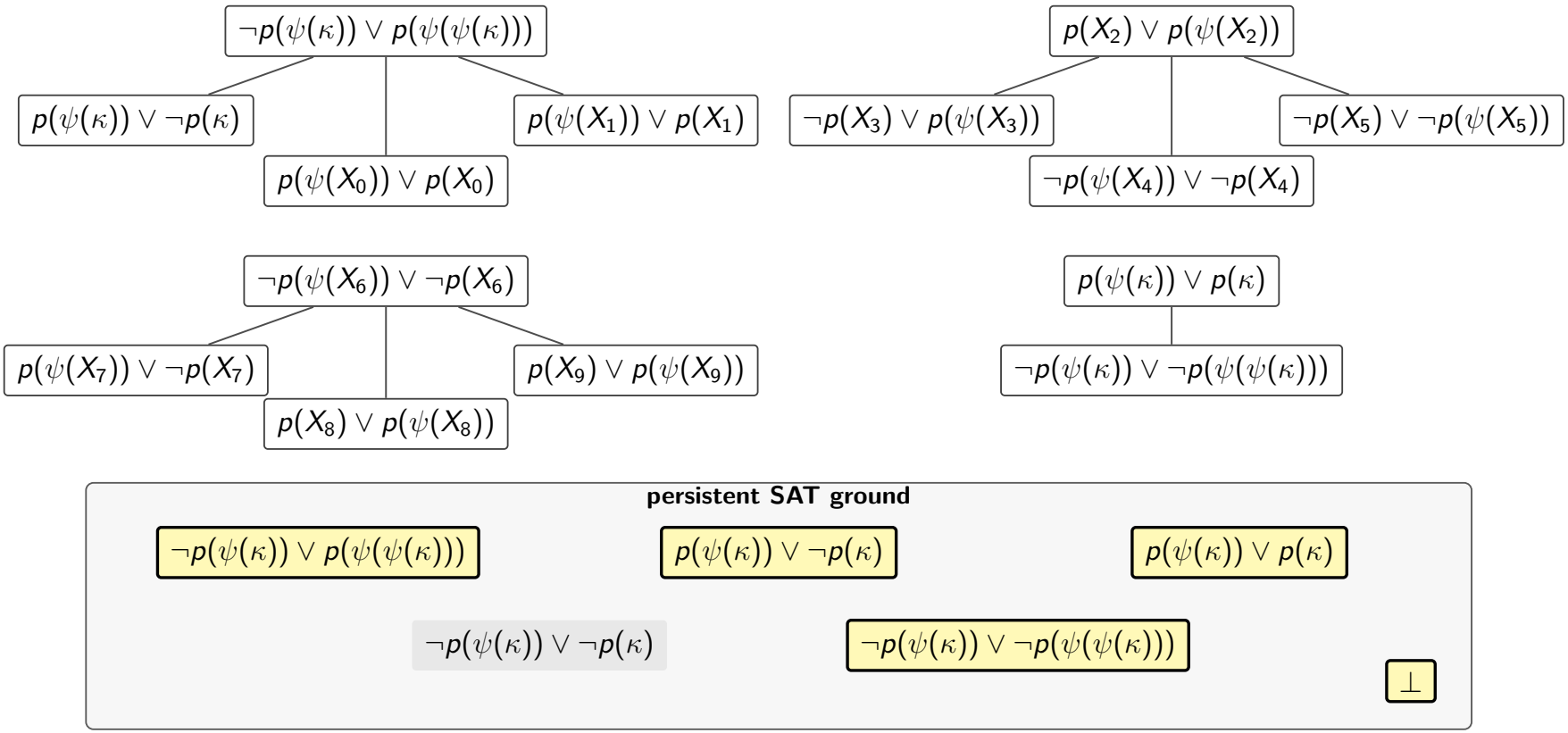


persistent SAT ground

$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$ $p(\psi(\kappa)) \vee \neg p(\kappa)$ $p(\psi(\kappa)) \vee p(\kappa)$

$\neg p(\psi(\kappa)) \vee \neg p(\kappa)$ $\neg p(\psi(\kappa)) \vee \neg p(\psi(\psi(\kappa)))$

SATCoP



total steps: 24

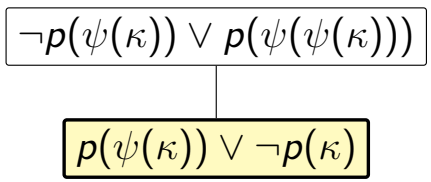
SATResetCoP

$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$

persistent SAT ground

$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$

SATResetCoP



persistent SAT ground

$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$

$p(\psi(\kappa)) \vee \neg p(\kappa)$

SATResetCoP



persistent SAT ground

$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$

$p(\psi(\kappa)) \vee \neg p(\kappa)$

$\neg p(\psi(\kappa)) \vee \neg p(\psi(\psi(\kappa)))$

SATResetCoP



persistent SAT ground

$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$

$p(\psi(\kappa)) \vee \neg p(\kappa)$

$\neg p(\psi(\kappa)) \vee \neg p(\psi(\psi(\kappa)))$

$p(\psi(\kappa)) \vee p(\kappa)$

SATResetCoP



persistent SAT ground

$\neg p(\psi(\kappa)) \vee p(\psi(\psi(\kappa)))$

$p(\psi(\kappa)) \vee \neg p(\kappa)$

$\neg p(\psi(\kappa)) \vee \neg p(\psi(\psi(\kappa)))$

$p(\psi(\kappa)) \vee p(\kappa)$

$\bot$

total steps: 5

Experimental Evaluation

Strategy	Implementation	Cut	SAT refutation	SAT guidance	**Reset** on **backtrack**
leanCoP	Prolog	✓			
SATCoP$_0$	Rust	✓	✓		
SATCoP	Rust	✓	✓	✓	
SATResetCoP	Python	✓	✓	✓	✓

Strategy	Implementation	Cut	SAT refutation	SAT guidance	**Reset** on **backtrack**
leanCoP^{con}	Python	✓			
SATCoP$_0$^{con}	Python	✓	✓		
SATCoP^{con}	Python	✓	✓	✓	
SATResetCoP^{con}	Python	✓	✓	✓	✓

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Connections Library

Connections is a Python library for building connection-tableau provers. It supplies representation of connection tableaus.
We implement **leanCoP^{con}**, **SATCoP$_0{}^{con}$**, **SATCoP^{con}**, and **SATResetCoP** as different policies over this same backend.

CaDiCaL 2.0

All SAT-based policies use the same incremental solver to retain and test the accumulated ground clauses.

Cambridge CSD3

Ice Lake node

Intel Xeon Platinum 8368Q at 2.60 GHz

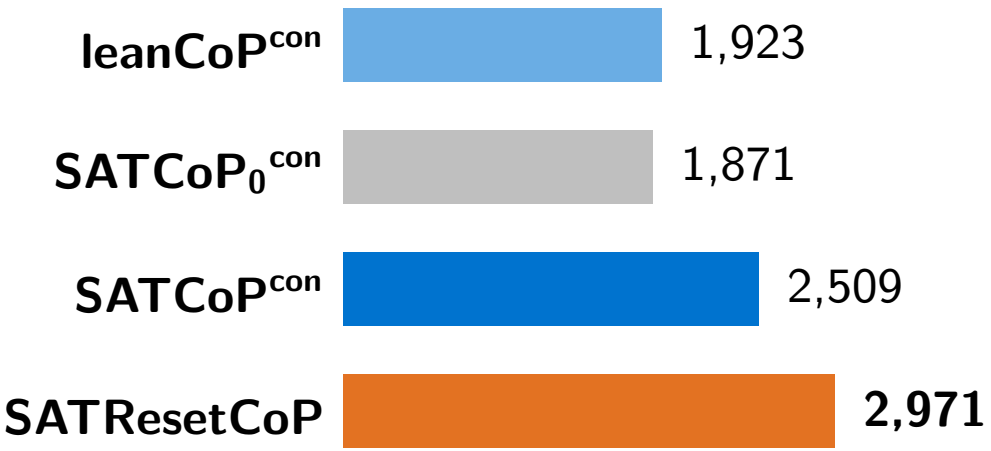
- dual socket, 38 cores each
- one hardware thread per core
- 3,370 MiB RAM per core
- each run uses one core

Rømming et al. (2023): *Connections: Markov Decision Processes for Classical, Intuitionistic and Modal Connection Calculi*. *Proceedings of the 1st International Workshop on Automated Reasoning with Connection Calculi*, CEUR Workshop Proceedings vol. 3613
Biere et al. (2024): *CaDiCaL 2.0*. *Computer Aided Verification*, Lecture Notes in Computer Science vol. 14681
University of Cambridge Research Computing Services (2026): *Ice Lake Nodes*

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Limits: 120 seconds or 1,000,000 steps per problem

TPTP problems solved



2,971
problems solved

+462
vs. **SATCoP (+18%)**

+1,048
vs. **leanCoP (+54%)**

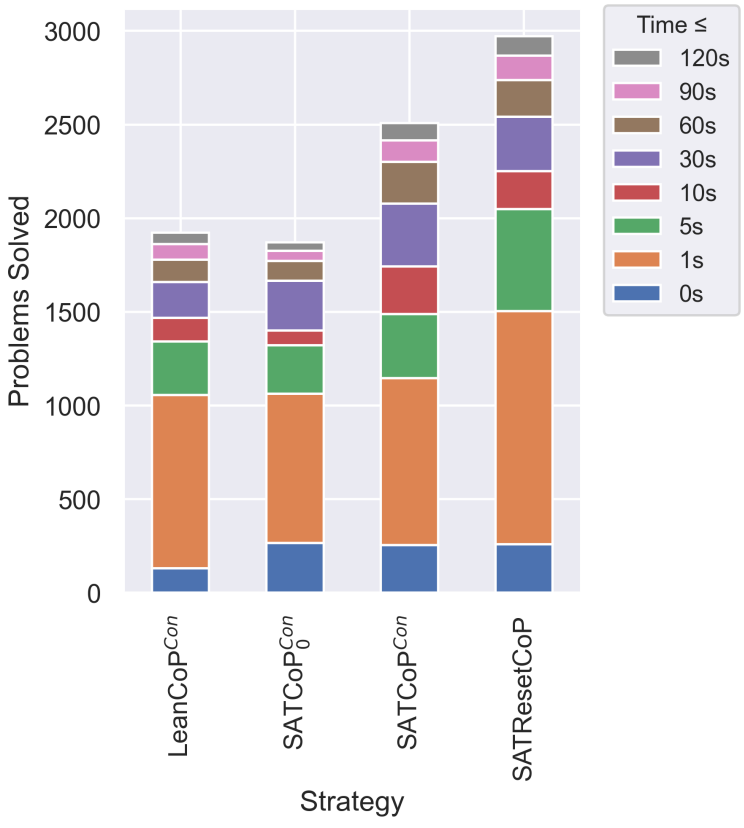
Aggressive resets solve 18% more TPTP problems than the best SATCoP baseline.

Sutcliffe (2017): *The TPTP Problem Library and Associated Infrastructure: From CNF to TH0, TPTP v6.4.0.* *Journal of Automated Reasoning* 59(4)

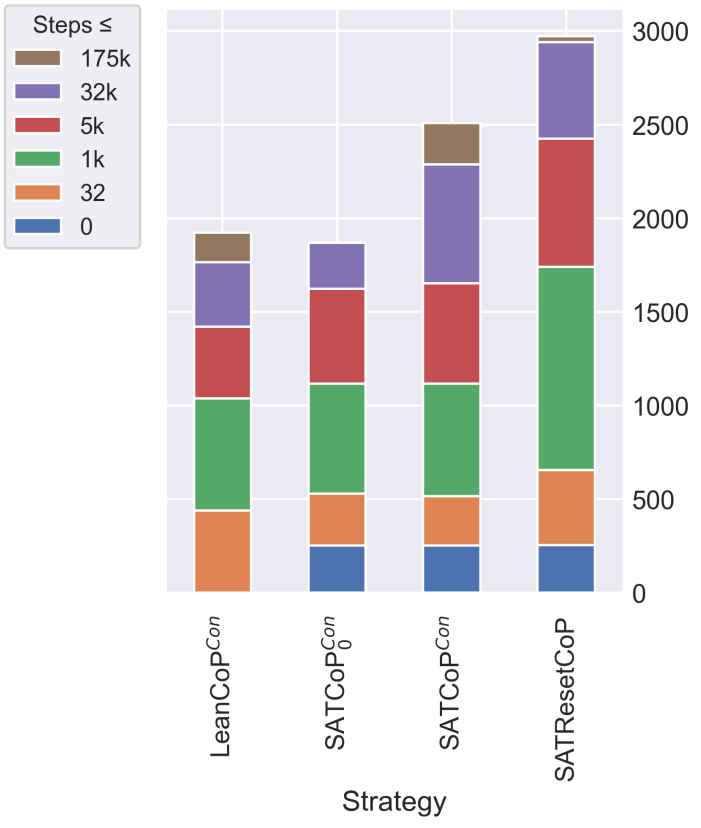
Faster Proving = More Proofs

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Solved by wall-clock time



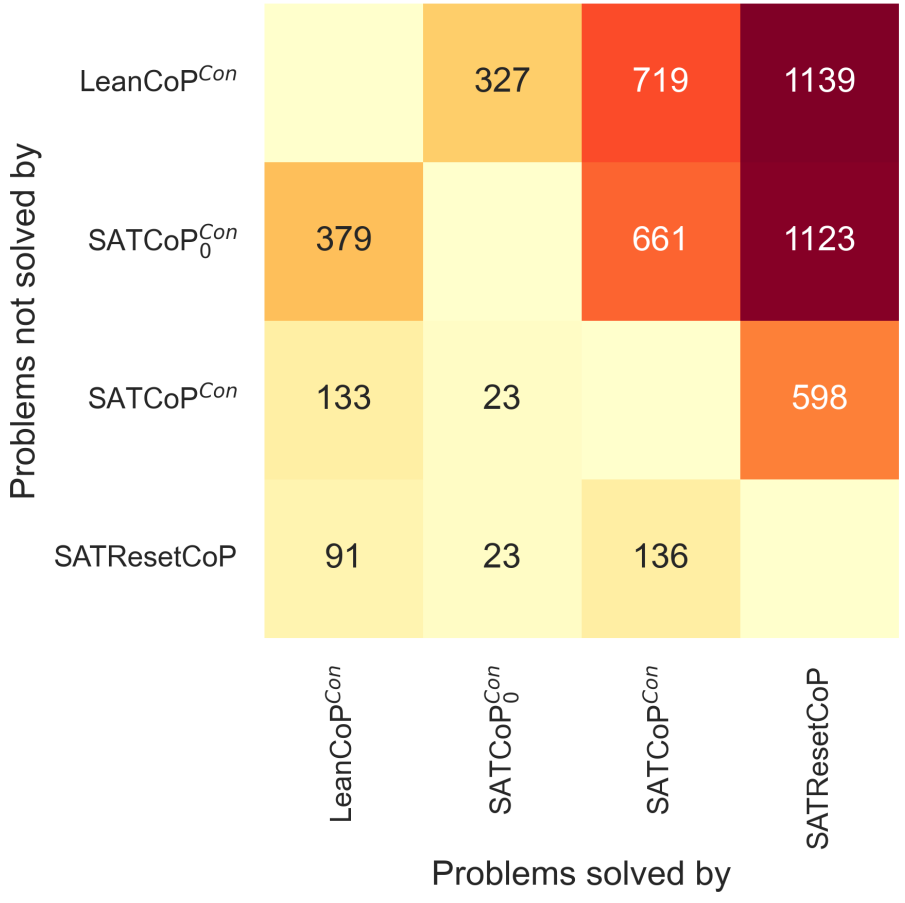
Solved by tableau steps



Resetting reaches more SAT refutations with less tableau search.

A 0-step proof means that the initial ground clause set was already propositionally unsatisfiable.

Resets Change the Search



Aggressive resets explore a different part of the search space—a schedule should retain both modes.

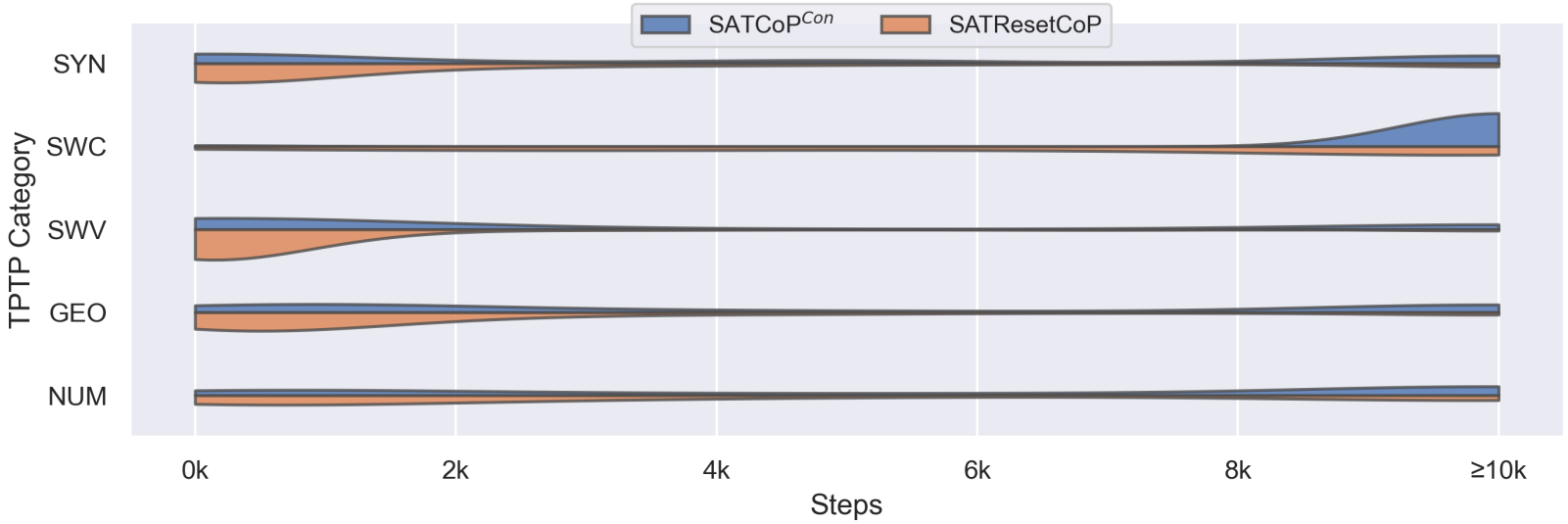
Where **Reset** Helps

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Top five absolute gains over SATCoP

	SATCoP	SATResetCoP	Gain
SYN	549	672	**+123**
SWC	11	64	**+53**
SWV	173	215	**+42**
GEO	260	293	**+33**
NUM	151	179	**+28**

Useful ground instances appear early, so retained clauses reach propositional **UNSAT** before local tableau search pays off.



Step distributions on the five categories above; unsolved problems are placed in $\geq$ 10k.

Takeaways

Keep the clauses, reset the tree.

- Aggressive resets help, but change which problems are solved.
- Schedule **Reset** search before conventional **SATCoP** or **leanCoP** search.


Mangla et al. (2022): *Bayesian Ranking for Strategy Scheduling in Automated Theorem Provers*. *International Joint Conference on Automated Reasoning (IJCAR)*
Biere et al. (2024): *CaDiCaL 2.0*. *Computer Aided Verification*, Lecture Notes in Computer Science vol. 14681

Keep the clauses, reset the tree.

- Aggressive resets help, but change which problems are solved.
- Schedule **Reset** search before conventional **SATCoP** or **leanCoP** search.
- **Try this at CASC this Monday!**


Mangla et al. (2022): *Bayesian Ranking for Strategy Scheduling in Automated Theorem Provers*. *International Joint Conference on Automated Reasoning (IJCAR)*
Biere et al. (2024): *CaDiCaL 2.0*. *Computer Aided Verification*, Lecture Notes in Computer Science vol. 14681

Acknowledgements

Thanks to Fredrik Rømming for his prior work on the Connections Prover.

This work was performed using resources provided by the Cambridge Service for Data Driven Discovery (CSD3) operated by the University of Cambridge Research Computing Service (`https://www.csd3.cam.ac.uk`), provided by Dell EMC and Intel using Tier-2 funding from the Engineering and Physical Sciences Research Council (capital grant EP/T022159/1), and DiRAC funding from the Science and Technology Facilities Council (`https://www.dirac.ac.uk`).

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Any questions?